

Module 9: Urinary system and toxicology

(Histology part)

• Functional histology of kidney, urinary tract and bladder

Function of kidney

- Filtration (by glomeruli), selective resorption and excretion (by tubular system)
- Endocrine function:
 - Fibrocyte in cortex=> erythropoietin
 - Modified fibrocyte in medulla=> prostaglandin to decrease BP

Histology of kidney

- 130-150 gm
- Outer granular cortex and inner striated medulla
- The medulla contains 10-18 medullary pyramids. The cortical tissue in between them is known as renal column of bertini
- Kidney is divided into lobes and lobules
- Renal lobe- one medullary pyramid and its corresponding area of cortical tissue
- Renal lobule-renal tissue surrounding a medullary ray

Basic structures

- **Uriniferous tubules:** functional unit of kidney; consist of:-
 - **Nephron** - active secretory part, 1-2 million in each kidney
 - **Collecting duct** - excretory part

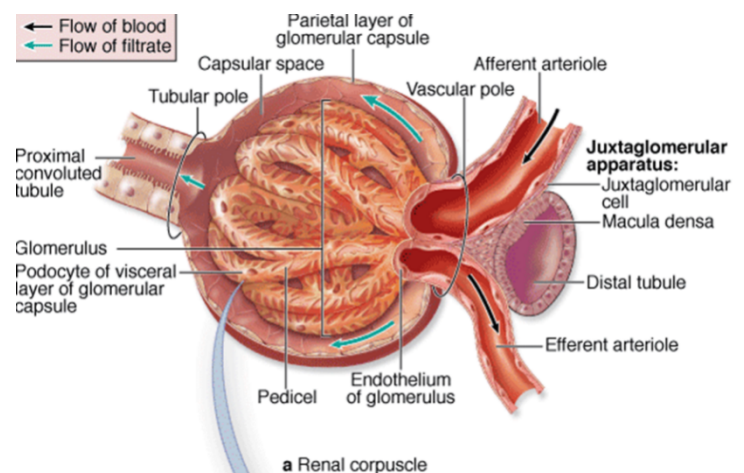
Nephron

1. Renal Malpighian corpuscle

- Found in the cortex.
- It consists of **bowman capsule** and **glomerulus**
- Function - formation of glomerular filtrate

a. Bowman capsule

- Double layered - **inner visceral (podocytes)** and **outer parietal** separated by **capsular space (urinary space)**.
- Podocyte- modified squamous epithelium that form non continuous covering of glomerulus capsule. These cells have major primary process and minor secondary process arising from the first.



b. Glomerulus

- Has blind beginning
- Has vascular and urinary pole
- It is tuft of capillaries whose endothelium is fenestrated but the basement membrane is thick and continuous.

- **Mesangial cells:** are faintly stained cells between blood capillaries

- Function- regeneration of basement membrane of endothelial cells.
- phagocytic, supportive and hormonal importance.

- control size of capillaries

- **Blood renal barrier-** separate blood from capsular space. It is formed of:

=>Endothelium of blood capillaries -

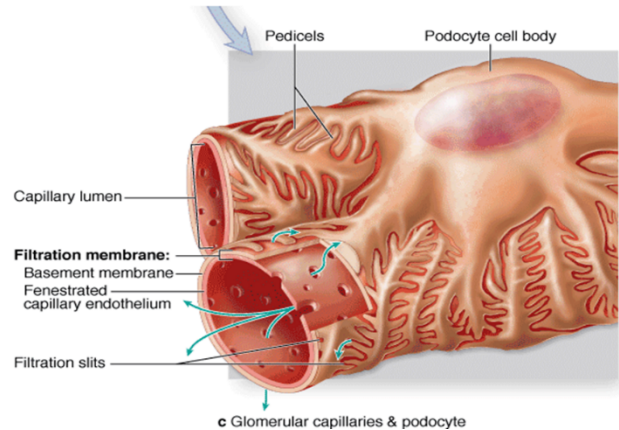
fenestrated with no diaphragm.

=>Basement membrane of endothelium-

- Thick; 3 layers:

- 2 laminae rarae – peripheral; for charge barrier
- central lamina densa - which contain collagen type 4 and act as physical barrier

=>Filtration slit- found between minor processes. It is covered by thin membrane or diaphragm.



2. Tubular system

- It consists of proximal convoluted tubules, loop of henle and distal convoluted.
- Depicted in the table starting from the next page.

Region	Epithelium	Function	Summary
Proximal convoluted tubule	Simple cuboidal epithelium with brush border; many compartmentalized mitochondria	Resorbs all glucose, amino acids, filtered proteins; at least 80% Na, Cl, H ₂ O	The Na pumps in basolateral membranes, transporting Na out of tubule, reduces volume of ultrafiltrate, maintains its isotonicity with blood
Loop of Henle, descending thick limb	simple cuboidal epithelium with brush border	Same as PCT	Same as PCT
Loop of Henle, descending thin limb	Simple squamous epithelium	Permeable to H ₂ O which exits ultrafiltrate and enters interstitium; Na, Cl enter ultrafiltrate	Ultrafiltrate becomes hypertonic with respect to blood; urea, from interstitium, also enters lumen of tubule

Loop of Henle, ascending thin limb	Simple squamous epithelium	Somewhat permeable to H ₂ O which enters ultrafiltrate from interstitium; Na, Cl exit ultrafiltrate	Ultrafiltrate remains hypertonic with respect to blood; urea, from interstitium, also enters lumen of tubule
Loop of Henle, ascending thick limb	Simple cuboidal epithelium; compartmentalized mitochondria	Impermeable to H ₂ O; Cl actively transported out of tubule into interstitium; Na follows	hypotonic Ultrafiltrate ;Cl pump in basolateral membranes is primarily responsible for establishing osmotic gradient in interstitium of outer medulla
JG apparatus macula densa	Simple cuboidal epithelium	Monitors level of Na (or decrease of fluid volume) in ultrafiltrate of distal tubule	Macula densa cells communicate with JG cells in Afferent arteriole via gap junctions
JG cells in afferent arteriole	Modified smooth muscle cells containing renin granules	Cells synthesize renin, release it into bloodstream	Renin acts on plasma protein, to trigger events leading to formation of angiotensin II, release of aldosterone from

			adrenal
Distal convoluted tubule	Simple cuboidal cells; compartmentalized mitochondria	Cells respond to aldosterone by removing Na from ultrafiltrate	Ultrafiltrate more hypotonic in presence of aldosterone; K, NH ₄ ⁺ , H ₂ O
Collecting tubules	Simple cuboidal epithelium; simple columnar epithelium	In absence of ADH, tubule impermeable to H ₂ O; hypotonic urine excreted	In presence of ADH, tubule permeable to H ₂ O, which is removed from filtrate, producing hypertonic urine

- **There are two types of nephrons:**

1. Cortical nephrons
2. Juxtamedullary nephrons

Cortical nephron	Juxtamedullary nephron
About 6/7 of the nephron	About 1/7 of the nephron
Have a short thin descending limbs	Short thick descending limb Long thin descending and ascending limbs

Short thin ascending limb	Thick ascending limb is present
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Ureters and urinary bladder

- **The calyces**, renal pelvis, ureter, and bladder have the same basic histologic structure, with the walls becoming gradually thicker closer to the bladder.
- The mucosa of these organs is lined by unique stratified transitional epithelium or urothelium.

The urothelium

- is composed of the following three layers:
 - a single layer of small basal cells resting on a very thin basement membrane.
 - an intermediate region containing from one to several layers of more columnar cells.
 - a superficial layer of very large, polyhedral or bulbous cells called umbrella cells which are occasionally bi- or multinucleated and are highly differentiated to protect underlying cells against the cytotoxic effects of hypertonic urine.

Microscopic structure of the urinary bladder

- Mucosa:
 - transitional type of epithelium and loose connective tissue
- submucosal layer
- Muscularis => smooth (detrusor) muscle fibers poorly arranged in three layers, outer and inner longitudinal and middle circular (most distinctly at the neck of the bladder)
- Serosa (in the fundus)/adventitia
- All the urinary passages are covered externally by an adventitial layer, except for the upper part of the bladder which is covered by serous peritoneum.

Microscopic structure of the ureter

- The wall of the ureter is formed of the same layers as the urinary bladder, but there are some differences:
 - Narrow lumen with **mucosal folds**
 - Thin wall

- The upper 2/3 of the ureteric length contains only two layers of smooth muscle but in the lower third a middle third circular layer develops.
- The fourth ureteric layer is adventitia formed of fibroelastic connective tissue.

Male and female urethra

- **The male urethra:**

- It is about 20 cm long. It is responsible for the passage of both the urine and semen.
- Four main parts form the male urethra:

- Prostatic urethra - lined by transitional epithelium
- Membranous urethra - lined by pseudostratified columnar epithelium (2cm)
- Penile urethra - lined by stratified columnar epithelium (15cm)
- Terminal urethra - lined by stratified squamous epithelium non keratinized

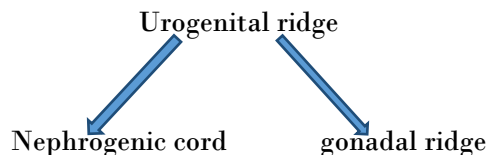
- **The female urethra**

- It is about 4-5 cm long responsible for the passage of the urine only.
- It is lined by transitional epithelium then stratified squamous non keratinized epithelium at its terminal part.
- Gland of Littre is found in the lamina propria of the female urethra, membranous and penile urethra of male.

(Embryology part)

• **Development and congenital malformations of kidney, urinary tract and bladder**

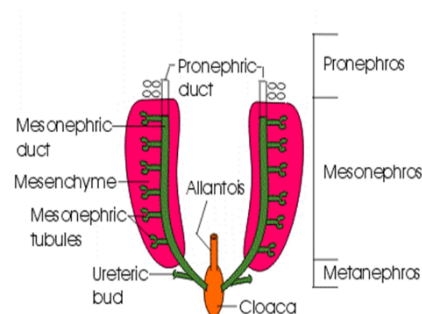
- Embryologically the urinary system and genital system are associated.
- Develop from intermediate mesenchyme in dorsal horn.



- Important genes=> wilms tumor suppressor (WT 1), steroidogenic factor 1, DAX1
- Mutations in these genes - x linked adrenal hypoplasia

Development of urinary system

- Before the gonadal system
- 3 set of kidney : -
 - **Pronephros** - 4 week, never functional, rudimentary, some



of its ducts are used by other sets

- **Mesonephrone** - late 4 week, function as interim kidney and degenerate but the mesonephric tubule become efferent ductules of testes

- **Metanephrone** - primordia of permanent kidney

- metanephric diverticulum--- collecting duct and ureter
- metanephrogenic blastema--- PCT, DCT differentiation.
- Reciprocal induction: one induces the formation of the other
- At term,
 - o All nephron formed.
- At 9th week
 - o Glomeruli filtration begins but functional maturation occurs after birth.
 - o Normal position attained. The kidney increases in size due to elongation of PCT and interstitium.
- As they ascend, they receive new branches from aorta and the caudal ones undergo involution.
- **Development of the Urinary Bladder:**
- cloaca is divided into a dorsal rectum and a ventral urogenital sinus by the urorectal septum.
- the urogenital sinus is divided into:
 - A cranial vesical part: most of the bladder and is continuous with the allantois but later constricts to give urachus which becomes median umbilical ligament.
 - A middle pelvic part: urethra in the bladder neck, the prostatic urethra in males, and the entire urethra in females.
 - A caudal phallic part: primordium of the penis or clitoris.
- The bladder develops mainly from the vesical part of the urogenital sinus, but its trigone region is derived from the caudal ends of the mesonephric ducts.
- Position of the developing bladder: abdomen in infants----greater pelvis at 6 yrs----lesser pelvis after puberty.
- **Development of the Urethra:**
- The epithelium of most of the male urethra and the entire female urethra is derived from endoderm of the urogenital sinus.
- In males, the distal part of the urethra in the glans of the penis is derived from a solid cord of ectodermal cells.
- The terminal part of the urethra is derived from the surface ectoderm.
- The connective tissue and smooth muscle of the urethra in both sexes are derived from splanchnic mesenchyme.

Accessory Renal Arteries

- Approximately 25% of adult kidneys have two to four renal arteries.
- Accessory (supernumerary) renal arteries usually arise from the aorta superior or inferior to the main renal artery and follow the main renal artery to the hilum of the kidney.
- An accessory artery to the inferior pole (polar renal artery) may cross anterior to the ureter and obstruct it, causing hydronephrosis-distention of the renal pelvis and calices with urine.
- If the artery enters the inferior pole of the right kidney, it usually crosses anterior to the inferior vena cava and ureter.
- It is important to be aware that accessory renal arteries are end arteries; consequently, if an accessory artery is damaged or ligated, the part of the kidney supplied by it will become ischemic.
- Accessory arteries are approximately twice as common as accessory veins.

Congenital Anomalies of Kidneys and Ureters

- Some type of abnormality of the kidneys and ureters occurs in 3% to 4% of newborn infants.
- Anomalies in shape and position are most common.
- Many fetal urinary tract abnormalities can be detected before birth by ultrasonography.

1. Renal Agenesis

- **Unilateral renal agenesis:**
 - occurs approximately once in every 1000 newborn infants.
 - Males are affected more often than females, and the left kidney is usually the one that is absent
 - Unilateral renal agenesis often causes no symptoms and is usually not discovered during infancy because the other kidney usually undergoes compensatory hypertrophy and performs the function of the missing kidney.
 - Unilateral renal agenesis should be suspected in infants with a single umbilical artery.
- **Bilateral renal agenesis**
 - is associated with *oligohydramnios* (small amount of amniotic fluid) because little or no urine is excreted into the amniotic cavity.
 - This condition occurs approximately once in 3000 births, and is incompatible with postnatal life because of the associated pulmonary hypoplasia.
 - These infants have a characteristic facial appearance:
 - the eyes are widely separated and have epicanthic folds
 - the ears are low-set
 - the nose is broad and flat
 - the chin is receding, and there are limb defects.
 - Most infants with bilateral renal agenesis die shortly after birth or during the first months of life.
- **Renal agenesis results:**

- when the metanephric diverticula fail to develop or the primordia of the ureters degenerate.
- Failure of the metanephric diverticula to penetrate the metanephrogenic blastema results in failure of kidney development because no nephrons are induced by the collecting tubules to develop from the metanephrogenic blastema.
- Renal agenesis probably has a multifactorial etiology.

2. Malrotated Kidney:

- If a kidney fails to rotate, the hilum faces anteriorly, that is, the fetal kidney retains its embryonic position.
- If the hilum faces posteriorly, rotation of the kidney proceeded too far.
- if it faces laterally, lateral instead of medial rotation occurred.
- Abnormal rotation of the kidneys is often associated with ectopic kidneys.

3. Ectopic Kidneys:

- One or both kidneys may be in an abnormal position.
- Most ectopic kidneys are located in the pelvis, but some lie in the inferior part of the abdomen.
- Pelvic kidneys and other forms of ectopia result from failure of the kidneys to alter position during embryo growth.

4. Horseshoe Kidney

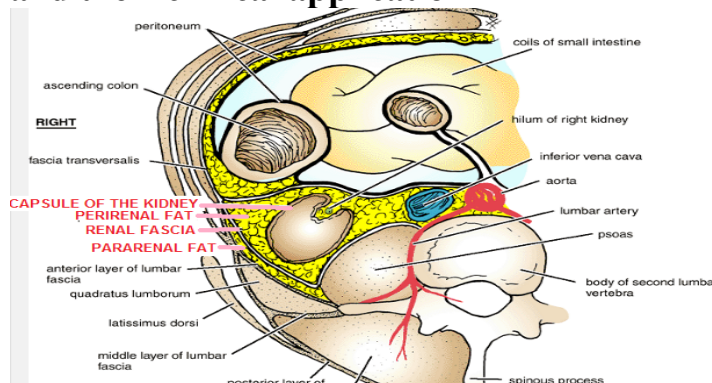
- In 0.2% of the population, the poles of the kidneys are fused
- Usually the inferior poles fuse.
- The large U-shaped kidney usually lies in the hypogastrium, anterior to the inferior lumbar vertebrae

5. Duplications of the Urinary Tract:

- Duplications of the abdominal part of the ureter and the renal pelvis are common.
- These anomalies result from division of the metanephric diverticulum.
- An ectopic ureter results when the ureter is not incorporated into the trigone in the posterior part of the urinary bladder.

(Anatomy part)

- **Important Relations and soft tissue coverings of the kidneys and ureter and their clinical application**



- Both the kidneys and associated adrenal glands are surrounded by varying degrees of fat and fascia:

 1. **The fibrous capsule** – a loose membrane which closely invests the kidney
 2. **Perirenal or perinephric fat** - layer of adipose tissue. It is thickest at the borders of the kidneys and fills up the extra space in the renal sinus.
 3. **Renal fascia (Gerota's fascia)**
 4. **Pararenal or paranephric fat** –more abundant posteriorly and towards the lower pole of the kidneys.

Gerota's fascia

- The anterior and posterior sheaths of Gerota's fascia become fused on three sides around the kidney (laterally, medially, superiorly)
 - Superiorly – fuse with the diaphragm(as a result kidney moves with respiration)
 - Medially – cross the mid line and fuses with Gerota's fascia of the opposite side.
 - Inferiorly – remains an open potential space, containing ureter and gonadal vessels.
- Gerota's fascia – forms an important anatomical barrier and tends to confine pathological process originating from the kidney. However, because of its deficiency inferiorly, a collection within it may track down and extend into the pelvis.

Surgical Significance of the Renal Fascia

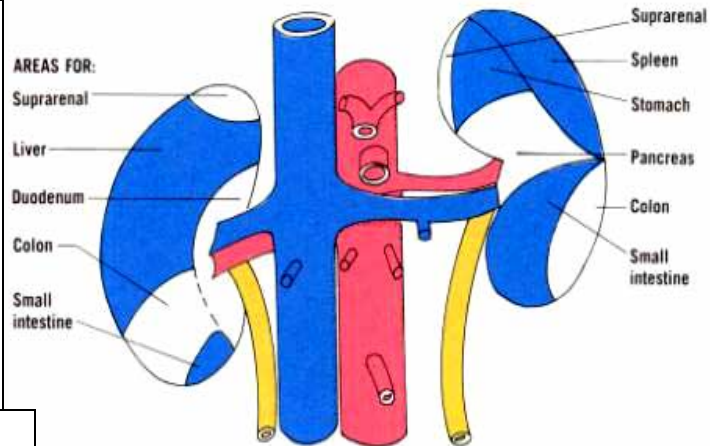
- The suprarenal glands, together with the kidneys, are enclosed within the renal fascia; the suprarenal glands, however, lie in a separate compartment, which allows the two organs to be separated easily at operation.

Relations of the kidneys:

- Anteriorly – not the same for the right and left kidneys.
- Refer next page

Anterior relations of right kidney –

- 1) **Below** – hepatic flexure of the colon + the small intestine (separated from it by peritoneum)
- 2) **Above** – supra renal gland
- 3) **Medially** – 2nd part of duodenum
- 4) **Laterally** – liver (separated from it by peritoneum)



Anterior relations of the kidneys

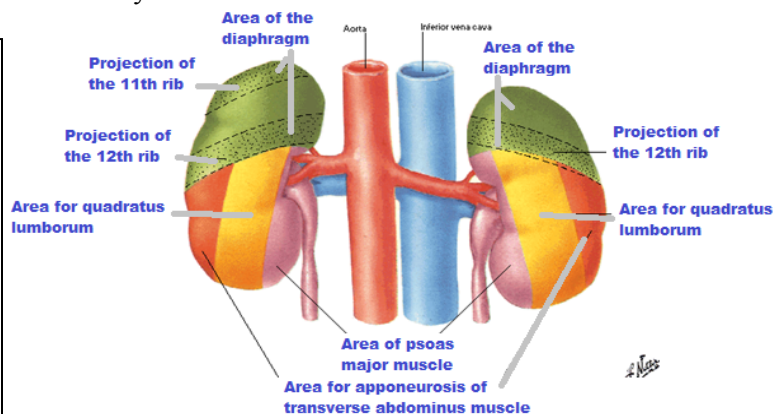
The areas covered by peritoneum are shown in blue.

Anterior relations of the left kidney

- 1) **Below and lateral** – splenic flexure of the colon
- 2) **Below and medial** – small intestine (separated by peritoneum)
- 3) **Middle and lateral** – pancreas
- 4) **Above and lateral** – spleen (separated by peritoneum)
- 5) **Above and medial** – stomach (separated by peritoneum)

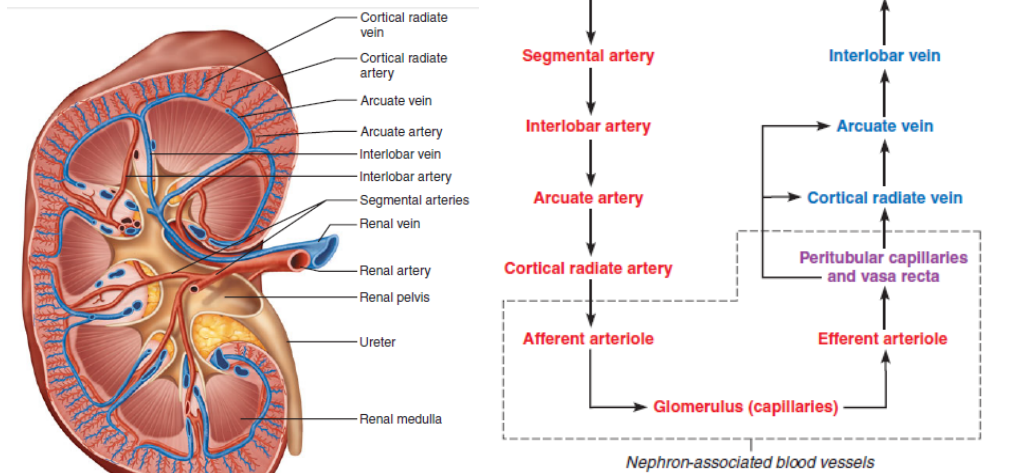
- Posterior relations – same in both kidneys

- 1) Psoas muscles
- 2) Transverse abdominus
- 3) Quadratus lumborum
- 4) Diaphragm
- 5) The subcostal vessels
- 6) The subcostal, iliohypogastric and ilioinguinal nerves.

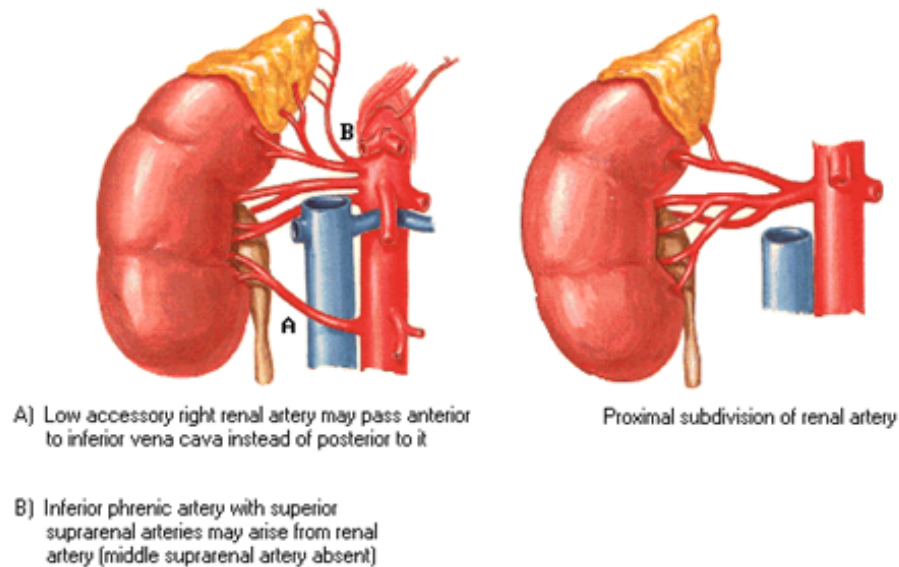


• Vasculature and lymphatics of the kidney

- More than 90% of the blood entering the kidney perfuses the cortex. These arteries give rise to the glomerular arterioles, which feed into the peritubular capillaries surrounding the tubules in the kidney.



Variations in Renal Artery



Lymph Drainage

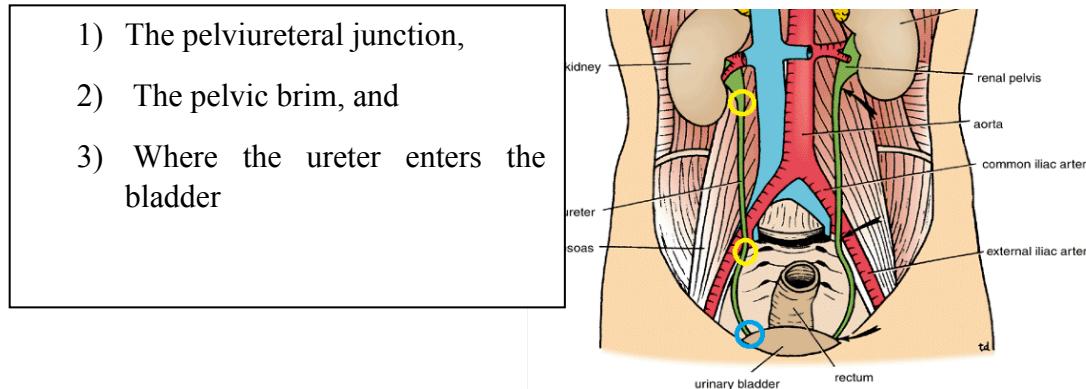
- Lymph drains to the lateral aortic lymph nodes around the origin of the renal artery.

Nerve Supply

- The nerve supply is the renal sympathetic plexus. The afferent fibers that travel through the renal plexus enter the spinal cord in the 10th, 11th, and 12th thoracic nerves.

- **Sites of anatomic narrowing of the ureter and their clinical applications**

- There are three sites of anatomic narrowing of the ureter where renal stones may be arrested, namely:



- **Renal pain, ureteric pain and referred pain and involved spinal nerves**

Renal Colic

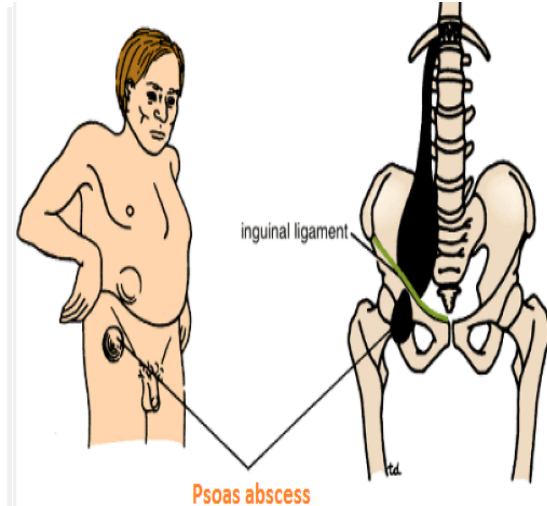
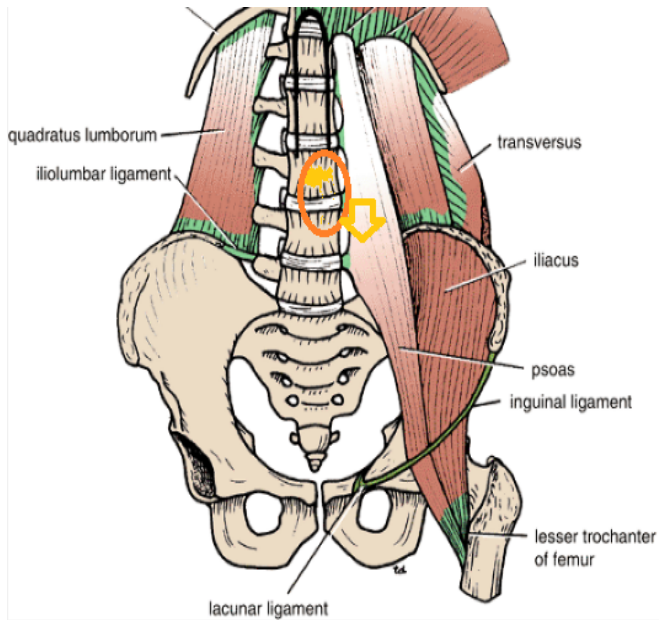
- The renal pelvis and the ureter send their afferent nerves into the spinal cord at segments T11 and T12 and L1 and 2.
- In renal colic, strong peristaltic waves of contraction to pass down the urine causes an agonizing colicky pain, which is referred to the skin areas that are supplied by these segments of the spinal cord, namely:
 - The flank
 - Loin
 - Groin
- When a stone enters the low part of the ureter(distal ureteric stone), the pain is felt at a lower level and is often referred to the testis or the tip of the penis in the male and the labium majus in the female.
- Sometimes ureteral pain is referred along the femoral branch of the genitofemoral nerve (L1 and 2) so that pain is experienced in the front of the thigh.
- The pain is often so severe that afferent pain impulses spread within the central nervous system, giving rise to nausea.

Psoas Fascia and Tuberculosis

- The psoas fascia covers the anterior surface of the psoas muscle and can influence the direction taken by a tuberculous abscess.
- Tuberculous disease of the thoracolumbar region of the vertebral column results in the destruction of the vertebral bodies, with possible extension of pus laterally under the psoas fascia.
- From there, the pus tracks downward, following the course of the psoas muscle, and appears as a swelling in the upper part of the thigh below the inguinal ligament. It may be mistaken for a femoral hernia.

Psoas sign

- Irritation of psoas muscle by inflamed appendix cause spasm and contraction of the psoas muscle => flexion of the thigh => pain in attempt to flex the thigh (at the hip joint)



Case of advanced tuberculous disease of the thoracolumbar region of the vertebral column. A psoas abscess is present, and swellings occur in the right groin above and below the right inguinal ligament.

(Physiology part)

• Kidney Function Tests (renal clearance)

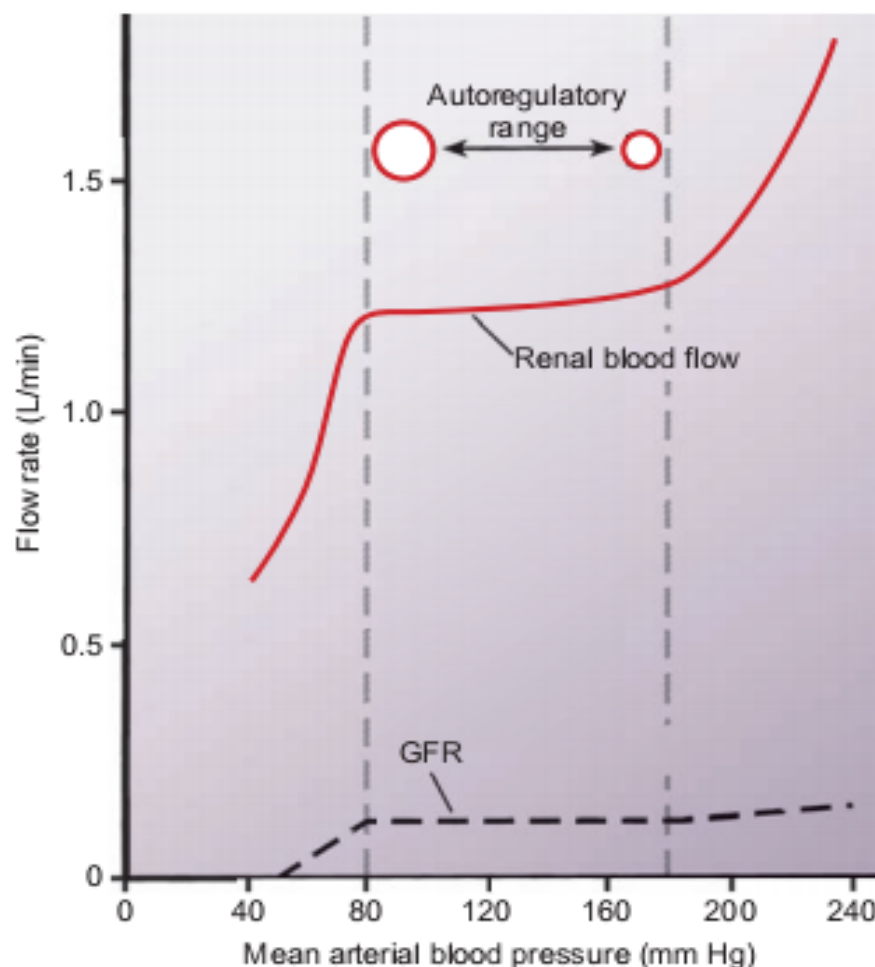
Use of Clearance Methods to Quantify Kidney Function

- Renal Clearance Is the Volume of Plasma That Is Completely Cleared of a Substance Each Minute.
 - o For a given substance X, $C_x(\text{ml/min}) = (U_x * V)/P_x$
 - Where; U_x = Conc. of a substance (X) in urine (mg/ml)
 V_x = Urine volume (ml/min)
 P_x = the conc. of the substance (X) in plasma (mg/ml)
- Renal clearances can be used to quantify several aspects of kidney functions, including the rates of glomerular filtration, tubular reabsorption, and tubular secretion of various substances.
- To understand the clearance concept, consider first substances such as inulin and Creatinine that are neither reabsorbed nor secreted by the nephron (i.e. their clearance is almost equal to GFR)
- Renal Clearance of Creatinine or Inulin Can Be Used to Estimate the GFR = 125ml/min
 - o A clearance value of less than 125 ml/min indicates reabsorption
 - o Greater than 125ml/min indicates secretion.

- NB: Cr clearance overestimates GFR by about 10% because there is some secretion into the proximal tubule.
- Glucose Clearance: Glucose like inulin is freely filtered, but all the glucose filtered is reabsorbed. Therefore, No plasma ends up losing glucose. Thus the clearance of glucose is zero (0).
- Clearance is a volume of plasma “cleaned” from a substance per unit time (mL/min), not the amount of solute excreted in urine per unit time (mg/min). Considering this, a clearance of 25 ml/min means that 25 ml of plasma is cleared from urea that is lost in urine every minute

- **Autoregulation: Tubuloglomerular feedback and myogenic mechanisms**

- Autoregulation is an intrinsic (local) property of the kidney by which GFR and Renal blood flow remains more or less constant or stable in conditions of wide changes in blood pressure.
- Such constancy of GFR & RBF helps H₂O and solutes not to escape through the urine by increased pressure changes.
- In the figure below, blood pressure keeps on increasing from ~80-to-200 mm Hg, but GFR and RBF show a flat curve (above) indicating no great changes in filtration or blood flow in the kidney.



✓ Two mechanisms of Auto-regulation

1. Myogenic (stress relaxation)
2. Tubuloglomerular feed back

1. Myogenic

- Increase in ABP causes the afferent arteriole to stretch greatly at the beginning. After a while, the stretch causes smooth muscles of arterioles to contract strongly.
- The increased stretch causes increased Ca^{2+} entry into smooth muscles of arterioles that in turn induce heavy constriction of the arteriolar walls.
- The vasoconstriction results in vascular resistance that decreases renal blood flow and decreased GFR
- If ABP decreases, the tonic constriction of arterioles disappears and the arterioles are dilated.
- Consequently, when mean ABP drops below 80 mm Hg, GFR decreases and less plasma is filtered to the Bowman's capsule. Less plasma filtration means that the potential for fluid loss through the urine is decreased.
- In other words, a decrease in GFR helps the body to conserve blood volume and thereby increase ABP.
- Vasodilatation, is however, not as effective as vasoconstriction in controlling ABP

2. Tubuloglomerular feed back

- Is a mechanism in which the composition of fluid in the early distal tubule is detected by the macula densa, which is located where the tubule abuts the afferent and efferent arterioles at the juxtaglomerular apparatus.
- An increase in GFR due to increase in ABP → Increased NaCl delivery to the macula densa → increased NaCl re-absorption → ATP of the macula densa cells are metabolized to produce adenosine that then combines with receptors in the afferent arteriole and cause vasoconstriction → lower blood flow in turn causes GFR to decrease to relatively normal value.

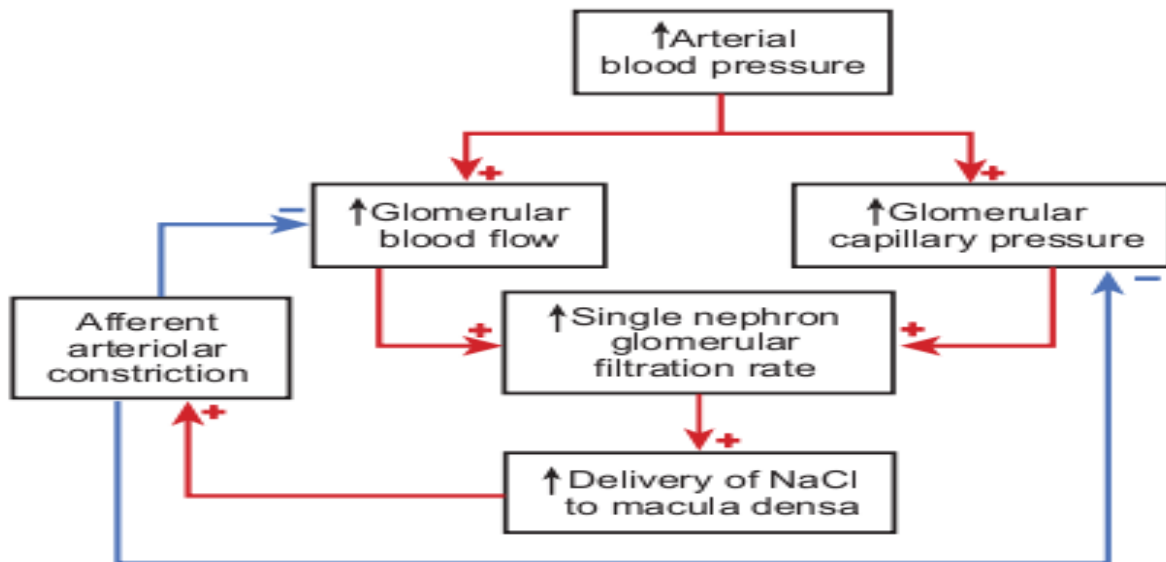
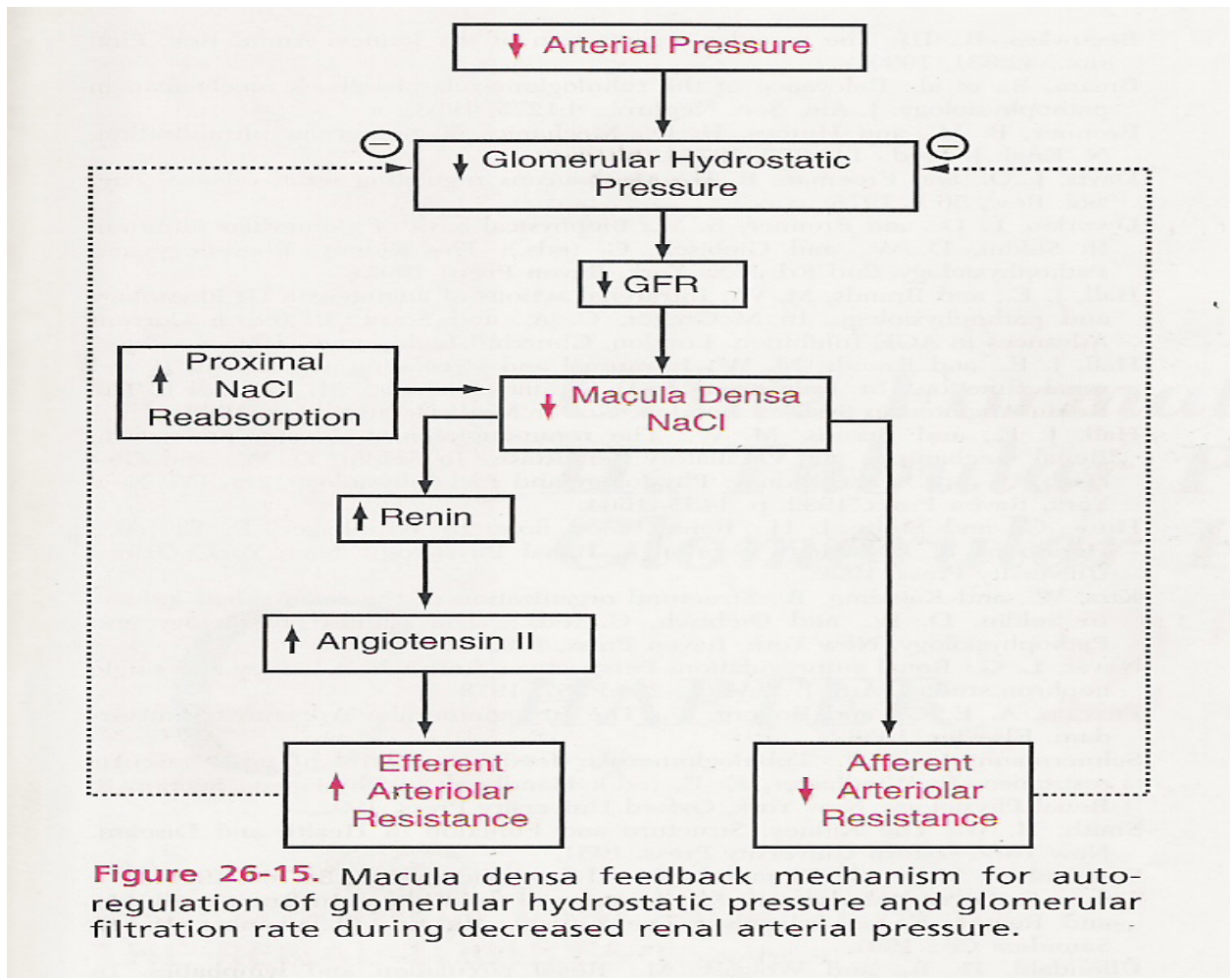


Figure 22.12 The tubuloglomerular feedback mechanism. When single-nephron glomerular filtration rate (GFR) is increased—for example, because of an increase in arterial blood pressure—more NaCl is delivered to and reabsorbed by the macula densa, leading to constriction of the nearby afferent arteriole. This negative-feedback system plays a role in auto-regulation of renal blood flow and GFR.

- Low ABP → Decreases GFR → decreases NaCl in the filtrate → causes low NaCl delivery to Macula Densa cells → Paracrine connection to JG-Cells that finally release Renin → Renin through RAAS produce Angiotensin II which causes efferent arteriolar constriction → this constrictor effect results in increasing GFR.



- → If a person takes protein rich diet → there'll be more amino acids in the kidney → increased Na⁺-AA co-transport → decreased NaCl delivery to macula densa cells → RAAS system activated → increases GFR
- → If there is heavy metal toxicity (Hg for e.g.) or if tetracycline damages the proximal convoluted tubule → inc. NaCl delivery to macula densa cells → decreases GFR

• Micturition reflex and Control of micturition: abnormalities

- The normal amount of Urine in bladder is about 400-500 ml. When it is full of urine, it stimulates stretch receptors → action potential through sensory neurons to → sacral spinal cord → reflex back through PSN to bladder → bladder is contracted → urine released through urethra. But at the same time, external urinary sphincter relaxes (through decreased motor signal from sacral).
- Pons and cerebrum also respond to micturition reflex. So, impulses ascend to micturition center, which finally causes modification of the reflex center
E.g. in children (2-3 years), the ability to voluntarily inhibit urination is not possible.

In adults, higher centers can inhibit PN to some degree.

- Spinal reflex, but also regulated by higher centers.
- PNS (pelvic nerves) innervates the bladder. SNS is not important.
- Stretch receptors are stimulated when the bladder is filled (~500 ml).
- Somatic fibers innervate the external sphincter (motor skeletal muscle voluntary).
- As the bladder continues to fill, micturition reflexes become more frequent and cause greater contractions of the detrusor muscles.
- When one micturition contractions start, it is Self-regenerative (stretch receptors are activated to a greater degree, to cause greater increase in sensory impulses to cause a strong degree of contraction).

• Diuresis

- A process in which urine is formed and it involves three steps 1) filtration 2) reabsorption and 3) secretion

Glomerular filtration

- Primarily depends on two factors: 1) Net filtration pressure= 10mmHg
2) Filtration coefficient (Kf)
- Net filtration pressure (10mmHg) = Favoring forces ($P_{GC} + \pi_{BC}$) - Opposing forces ($P_{BC} + \pi_{GC}$)
I.e. P_{GC} - hydrostatic pressure of the glomerular capillaries = 50mmHg
 π_{BC} -colloid osmotic pressure in Bowman's capsule = 0mmHg
 P_{BC} - hydrostatic pressure of the Bowman's capsule = 10mmHg
 π_{GC} - colloid osmotic pressure of the glomerular capillaries = 30mmHg
- The numerical values may vary in different texts but the Net filtration pressure is always the same=10mmHg.
- Filtration coefficient depends on two factors – permeability of the filtering membrane and total surface area of the filtering membrane.
- $GFR = NFP * Kf$, from this $Kf = 125 \text{ ml/min} / 10 \text{ mmHg} = 12.5$
- Filtration fraction: is defined as the fraction of renal plasma flow (RPF) that finally becomes Glomerular filtrate.

$$FF = \frac{GFR \text{ (125 ml/min)}}{RPF \text{ (i.e., ~650 ml/min)}} \times 100$$

$$= \frac{125}{650} \times 100 = 19\% - 20\% \text{ of the renal plasma flow accounts to the GFR.}$$

- π_{GC} can be affected by filtration fraction: increased FF $\rightarrow$ plasma remaining in the capillaries become highly concentrated $\rightarrow$ increased π_{GC} $\rightarrow$ decreased GFR
- Generally, Substances filtered are H_2O , major electrolytes (Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- ,), metabolic waste products (urea, creatinine), metabolites (glucose, a.a.s, ketone bodies), low molecular wt proteins (insulin, Hemoglobin- if free in plasma) and other non- natural substances like few drugs, inulin(which is used to measure GFR), Paraaminohippuric acid-PAH (marker of renal plasma flow).
- Blood cells aren't filtered, plasma proteins and plasma protein bound substances aren't also filtered (e.g. bilirubin, fatty acids, calcium, many drugs, T3, T4, etc)

Tubular processing of glomerular filtrate – Reabsorption

- 650ml of plasma per minute passes the renal circulation – 20% of this amount is filtered (125ml/min).
- 65% and lots of solute reabsorbed in Proximal Convolute Tubule (PCT)
- 10-15% in descending loop of Henle
- Under all physiological circumstances, no water is reabsorbed from ascending loop of Henle.
- 15% reabsorbed in Distal CT.
- ~5ml of filtrate reaches the collecting duct and ~4ml of it will be reabsorbed under the influence of aldosterone (reabsorbs salt and water, and excretes K^+) and ADH (water reabsorption).
- Thus, in healthy individuals 99% of the fluid that reaches the kidneys is reabsorbed excreting 1ml drop of urine per minute (by both kidneys together) – 500-3500ml/day of urine is excreted.
- Reabsorption occurs either transcellularly (within the cells) or paracellularly (through gaps in b/n the cells)

Urine excretion = glomerular filtration - tubular reabsorption + tubular secretion

PCT

- 100% of glucose, 65% of H_2O , 65% of Na , and 100% of amino acids get absorbed.
- Cells of PCT have apical brush borders which increase the surface area by 20X, and they also have tight junction's b/n cells that only allows water reabsorption.
- Basolateral side is concentrated by Na^+/K^+ ATPases ($3Na^+$ out and $2K^+$ in – making PCT cells Na^+ poor and more electronegative) and basal mitochondria which make the striations.
- Na^+ enter from tube to cells by facilitated diffusion, and glucose and amino acids are actively co-transported with Na^+ (2^o active transport) and basolaterally they move out of the cells through facilitated diffusion. Cl^- is also co-transported with sodium.
- Some solutes are also dragged with H_2O (i.e. solvent drag) chloride is one of them.
- In the lumen: $H^+ + HCO_3^- \rightarrow H_2CO_3 \rightarrow CA \rightarrow CO_2 + H_2O$ (in to the cells)
- Inside the cells: $H_2O + CO_2 \rightarrow CA \rightarrow H_2O_3 \rightarrow HCO_3^- + H^+$

- The H^+ is secreted out through by Na^+/H^+ antiport and the HCO_3^- is diffused out of the cells into the interstitium using facilitated diffusion.
- Very small proteins like hemoglobin and insulin can filter into the kidneys. Hemoglobin in the cells of PCT is catabolized and releases Fe which is toxic and damages the cells leading to tubular necrosis.
- Small proteins bind to the membrane and attract clathrin which uses ATP – pinocytosis.

Transport Maximum:

- Is the maximum rate (mg/min) at which substances can be completely reabsorbed or secreted in the kidney tubules. Factors on which T_m depends are Carrier proteins and Presence of enzymes.
- E.g. T_m for glucose = 320 mg/min. When tubular load is less than 320 mg/min, then all of the glucose will be reabsorbed.
- When there happens to be large increase in GFR or plasma glucose concentration that increases the filtered load (>320 mg/min), the excess is excreted as urine.
- All active transport systems have transport maximum (T_m) which implies a limit for the amount of the substance that can be transported /unit time.
- Renal threshold: is the filtered load at which that substance starts to appear in the urine (200- 300 mg/dl).

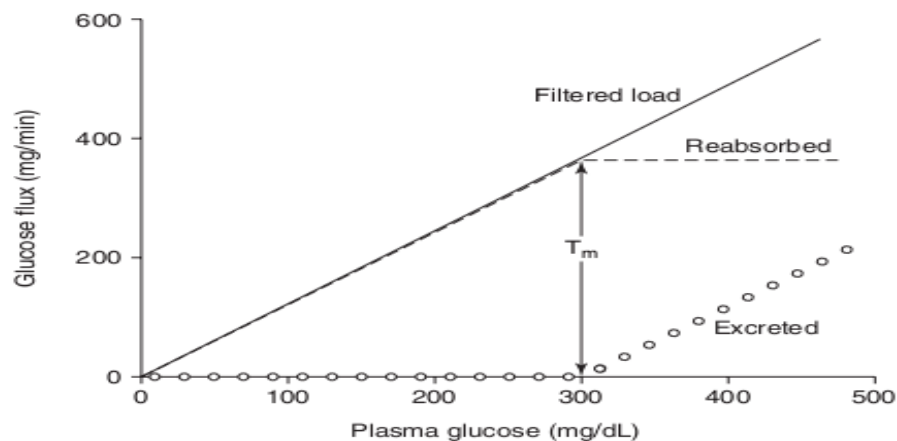


Figure 5–1. Glucose handling by the kidney. The solid line shows the filtered load. It is strictly proportional to the plasma concentration when the GFR is fixed at 125 mL/min. The reabsorption rate exactly matches the filtered load for loads less than the T_m of 375 mg/min. At all these loads, the excretion rate is zero. For filtered loads greater than the T_m , excretion becomes greater than zero and rises in equality with the difference between the filtered load and the T_m .

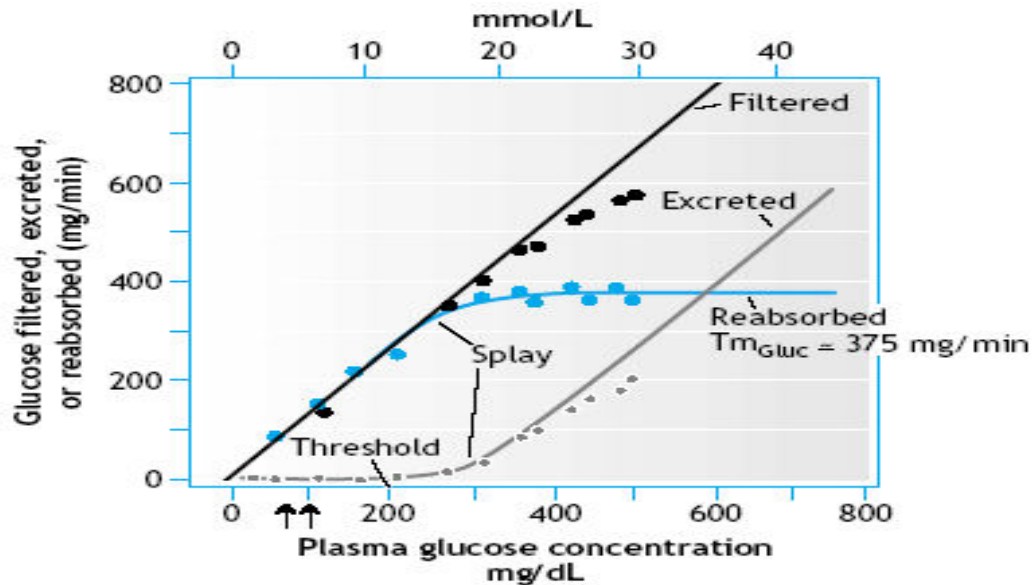


Figure 4-12. Renal handling of glucose. $T_{m_{gluc}}$ is the transport maximum or the maximal rate at which the renal transport system can reabsorb glucose. At higher concentrations, the $T_{m_{gluc}}$ is saturated and glucose begins to appear in the urine.

- The bending in reabsorption and excretion graphs known as splay indicates the heterogeneity of nephrons i.e. all nephrons do not have the same transport maximum.

Net effects of reabsorption and excretion

- Substances which are neither reabsorbed nor secreted are called inert substances. So, for inert substances, filtered load equals excretory load (there is no net tubular modification). E.g. inulin and creatinine.
- Net reabsorption: filtered load > excretory load.....e.g. glucose, a.a.s, Na^+ , Cl^- , etc
- Net secretion: filtered load < excretory load.....e.g. organic acids and bases, PAH, bile salts, oxalates, urates, catecholamines (VMA), toxins, drugs, etc
- Urea is also absorbed but in less amount.
- PO_4^{3-} absorption depends on parathyroid hormone – normally it doesn't allow reabsorption of PO_4^{3-}
- Angiotensin II increases reabsorption of Na^+ at PCT.
- α_1 - hydroxylation of vitamin D which is also stimulated by PTH.
- As we go down in PCT, the amount of Na^+ is decreasing but the concentration of Na^+ is constant.....thus making the lumen **isosmotic**.

Loop of henle

- **Descending limb:**
 - Thick part is similar to PCT.(medullary diluting units)
 - Thin part:

- Fluid in interstitium of medulla is hyperosmotic (more solute).
- Thin part of descending limb of loop of henle – no brush border, no much mitochondria, metabolically minimally active but freely permeable to water.
- More solute will be gained making it hyperosmolar.
- Considering water reabsorption:
 - PCT (65%) and descending limb(20%) – water permeable
 - Ascending limb and early DCT – water impermeable
 - Late DCT and collecting duct – permeability is regulated by ADH
- **Ascending limb**
- Thin part: also called diluting segment of the lumen and concentrating segment of the interstitium. The tight junction is water tight but not solute tight....so there will be much water in the lumen not absorbed making it dilute and at the same time solutes get absorbed making the interstitium concentrated. Normally 25% of Na^+ is absorbed here.
- **Thick part:**
- Basolateral membrane is rich in Na^+/K^+ ATPases and the apical membrane is rich in $\text{Na}^+/\text{K}^+/\text{Cl}^-$ co-transporters – site of blockage by loop diuretics.
- K^+ leaky channels are found both basolaterally and apically which contribute for making the luminal potential to be electropositive. This electro positivity pulls the ions Ca^{2+} and Mg^{2+} to get reabsorbed paracellularly. ('Like charges repel' principle).

Early DCT

- Basolateral membrane has Na^+/K^+ ATPases also Ca^{2+} ATPases and apical membrane has Na^+/Cl^- co-transporter.
- Ca^{2+} is also reabsorbed here because the $\text{Ca}^{2+}/\text{Na}^+$ exchanger and Ca^{2+} ATPases make the cells Ca^{2+} poor. PTH plays an important role in Ca^{2+} absorption at this site through activation of transmembrane G-proteins and facilitates absorption of Ca^{2+} .

Principal cells of late DCT and collecting ducts

- Are more abundant in collecting ducts.
- Basolateral membrane rich in Na^+/K^+ ATPases, and apical membrane consists K^+ and Na^+ channels which absorb sodium and secrete potassium.
- Aldosterone facilitates function of principal cells through binding to its cytoplasmic receptor, translocated into the nucleus, then activate genes for Na^+/K^+ ATPase production, and Na^+ and K^+ channels.
- Water channels are also there in the basolateral membrane called aquaporin 3&4. Aquaporin 2 is planted within the vesicle inside principal cells. ADH causes fusion of this vesicle with the membrane and appearance of aquaporin 2 for H_2O reabsorption.

Intercalated cells late DCT and collecting ducts

- Found intercalated between principal cells and are H^+ (acid) secreting cells by H^+ ATPases in apical membrane. Very rich in carbonic anhydrase.

- H^+ ATPases pump H^+ against 1000 fold concentration.
- Provide HCO_3^- to the interstitium through facilitated diffusion
- The secreted H^+ bind with HPO_4^- and form H_2PO_4 or with NH_3 and form NH_4

Medullary collecting ducts

- Some cells work like intercalated cells and some work like principal cells.
- This area is a unique site for having special transporters for urea absorption.
- The absorbed urea In the medullary interstitium go to loop of henle and come back luminally to the collecting duct and the cycle is repeated again and again.....**Urea Recycling**

Mechanisms of urine concentration and dilution

- Normal osmolality of the blood is 290-300 mosm/l. When the blood is hyperosmolar the kidney produces concentrated urine and when it is hypoosmolar the kidney produces diluted urine.
- The Hyperosmolar medullary interstitium is produced by counter current multiplier and urea recycling in renal medullary area, and it is maintained by vasa recta as a counter current exchanger.

ADH amplifies counter current multiplier and urea recycling: in vasa recta when blood is going down, it loses water and gain solutes to equilibrate with progressive hyperosmolar medullary interstitium. When blood ascends back to the cortex it loses solutes and gains water and by the time the blood leaves vasa recta, its osmolality is ~325mosm/l.....and this is called counter current exchanger.

Counter current multiplier

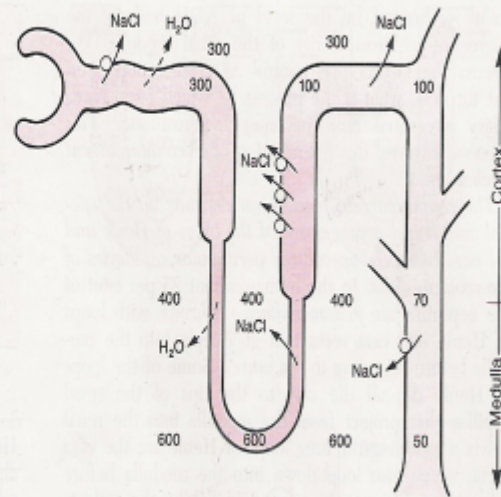
- **Loop of henle**
- **Active transport**
- **Generate hyperosmolarity**

Counter current exchanger

- **vasa recta**
- **passive transport**
- **for maintenance**

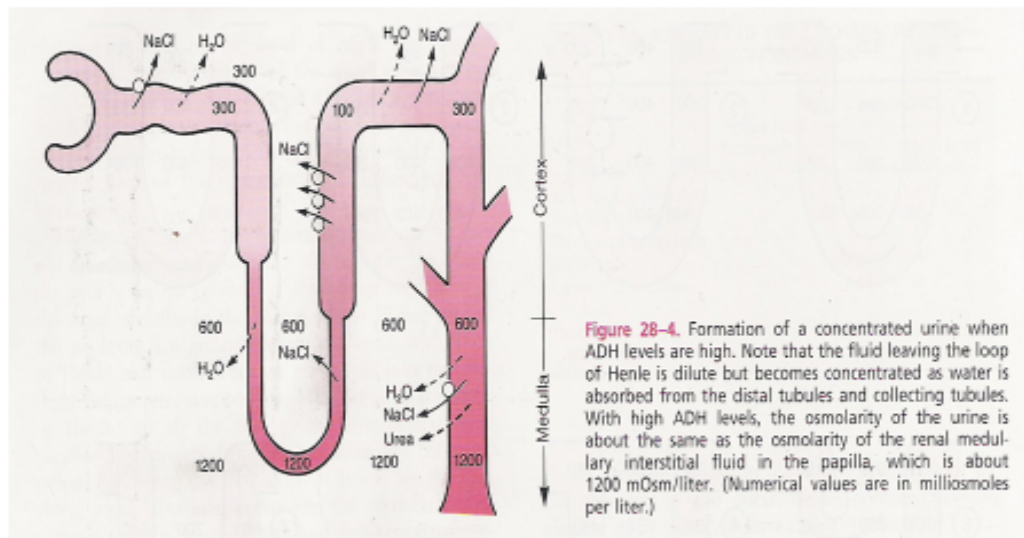
Formation of dilute urine when ADH is low

-2. Formation of a dilute urine when ADH is very low. Note that in the ascending loop of Henle tubular fluid becomes very dilute. In the distal and collecting tubules, the tubular fluid is diluted by the reabsorption of sodium chloride and failure to reabsorb water when ADH levels are low. The failure to reabsorb water and continued reabsorption of solutes lead to a large volume of dilute urine. (Numerical values are in milliosmoles per liter.)



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Formation of a concentrated urine, ADH is high



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- **Renal regulation of acid base balance**

- Kidneys regulate acid base balance via 3 fundamental mechanisms:
 - Secretion of H^+
 - Reabsorption of filtered HCO_3^-
 - Production of new HCO_3^-

- First let's see what happens in the PT (proximal tubule)

Apical side (toward the lumen)

1) H goes out to the filtrate and Na moves in to the tubule by a H/Na anti port.

This is derived by the concentration gradient formed at the basolateral side by Na/K ATPase.

- Since bicarbonate cannot be absorbed by the apical side of the lumen what happens is it reacts with the H that went out to form carbonic acid in the presence of brush border CA. This acid in turn dissociates and forms CO_2 and H_2O . so CO_2 and water easily diffuse to the lumen. Next these compounds unite in the presence of the intracellular CA to form carbonic acid which will dissociate to give H and HCO_3^- . HCO_3^- will diffuse out of the basolateral membrane through $Na-HCO_3$ and $Cl-HCO_3$ channels
- 2) In the DT (distal tubule) H is secreted by the $HATPase$ pump. This is responsible for the acidification of urine (PH-5-7)
- Secretion Happens in special type of cells called intercalated cells. What makes the DT different from other tubules is here the excretion of H is via H ATPase unlike the others which use the anti port system.
- Note that although PT and DT are mentioned all the other tubules except thin DLH and thin ALH are all involved in the secretion of H and reabsorption of HCO_3^- .
- $\uparrow PCO_2$, $\uparrow H^+$, $\downarrow HCO_3^-$, $\downarrow ECF$ volume, $\uparrow$ Angiotensin II, $\uparrow$ Aldosterone and Hypokalemia increase H secretion and HCO_3^- reabsorption. The reverse is true if these values are the opposite.

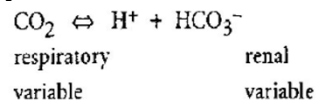
Acid base disturbances

- Respiratory acidosis: $\uparrow PCO_2$ of ECF
- Caused by decreased rate of ventilation, obstruction, medullary respiratory center damage.....
- Respiratory alkalosis: $\downarrow PCO_2$
- Usually caused by ascending to high altitude; low level of O_2 increases ventilation rate.
- Metabolic alkalosis: $\uparrow ECF [HCO_3^-]$
- Due to Diuretics (exc CA inhibitors), Hyperaldosteronism, Vomiting of gastric content, Ingestion of alkaline drugs
- Metabolic acidosis: $\downarrow ECF [HCO_3^-]$
- Due to – Renal tubular acidosis, Diarrhea, Vomiting from deeper GIT, Diabetes Mellitus, Ingestion of acids
- Disorders can be analyzed using 3 types of measurements ;
 - 1) PH-normal value = 7.4
 - If >7.45 = alkalosis
 - If <7.35 = acidosis
 - 2) plasma $[HCO_3^-]$ – normal = 24mEq/L

- If less than this value and there is a decrease in CO₂ to compensate that—metabolic acidosis
- If more than this value +increased CO₂ to compensate it—metabolic alkalosis
- 3) Pco₂:normal = 40mmHg
 - If more than this value + compensatory increase in HCO₃=respiratory acidosis
 - If less than this value+ compensatory decrease in HCO₃= respiratory alkalosis

Compensatory mechanisms

1) Respiratory : occurs in metabolic disturbances only



Metabolic acidosis: hyperventilation (shift to the left)

Metabolic alkalosis: hypoventilation (shift to the right)

2) Renal

- Takes part in both respiratory and metabolic disturbances
- Slower than the respiratory compensation

Acidosis: HCO₃ production by the kidney and H excretion.(acidic urine)

Alkalosis: Hco₃ excretion (alkali urine)

- For complex disturbances which involve both the respiratory and metabolic system we use an acid base nomogram.
- For e.g. if a pt with pH= 7.15, [HCO₃-]_p = 10 mEq/L, and Pco₂= 60 mm Hg
We should note that he is acidotic and since there is no compensation by the respiratory system + an increase in CO₂ value, we can conclude that both systems are affected.

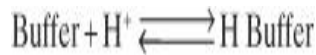
- The disorders can be diagnosed by analyzing: - the anion gap
- the urine pH

Anion gap

- In blood plasma the number of cations and anions is equal.
- Major cation => Na
- Major anion => Hco₃ and Cl
- The minor ones are called unmeasurable anions or cations.
- Anion Gap is the difference b/n [Na⁺] & Σ[HC03⁻] & [Cl⁻]. Here the unmeasurables are also included.
- Normal AG = 12 + or - 2
- Usually used to identify the cause of metabolic acidosis.
- So if there is an acidosis then there will be an increase in anion gap
- Largest gaps are caused by ketoacidosis and lactic acidosis
- One exception here is hyperchloremic acidosis-here the anion gap is normal because the rise in Cl compensates for the fall in HCO₃.

- **Chemical buffer system in acid-base homeostasis**

- A buffer is any substance that can reversibly bind to H^+ .it is all about keeping the PH close to the normal value which is 7.4.



- If there is an increase in H then it shifts to the right forming a weaker acid.
- If there is a decrease in H then it shifts to the left releasing H.
- So the buffer system minimizes change in PH or (H)
- Buffers can be extracellular or intracellular
- Extracellular:
 - The bicarbonate buffer system (major)
 - The phosphate buffer system (minor) (it is mentioned as intracellular in some references since its extracellular use is very negligible)
- Intracellular:
 - Organic phosphates [e.g., AMP, ADP, ATP, 2,3-diphosphoglycerate (DPG)]
 - Proteins
- We will take a look at the 3 main buffer systems which are

The bicarbonate buffer system

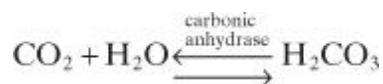
The phosphate buffer system

The Protein buffer system

- 1) **The bicarbonate buffer system**

- contains: **A)** H_2CO_3

which is formed by the reaction

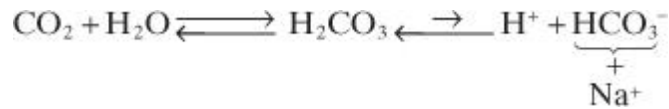


- This reaction is very slow and happens only in the presence of carbonic anhydrase (which is abundant in lung alveoli(where CO_2 is released),renal tubules, RBCs, Gastric mucosa, pancreatic cells)
- After being formed H_2CO_3 ionizes weakly to form small amounts of H^+ & HCO_3^-
- rxn: $H_2CO_3 \rightarrow H^+ + HCO_3^-$

- B) $NaHCO_3$**



- overall reaction:



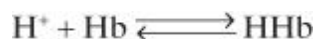
- What happens is when a strong acid like HCl is introduced to the ECF it is buffered by HCO_3^- and gives out H_2CO_3 , which will in turn dissociate to CO_2 and H_2O . here you should note that the released CO_2 is expelled through a rise in respiration.
- rxn: $\uparrow \text{H}^+ + \text{HCO}_3^- \longrightarrow \text{H}_2\text{CO}_3 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
- on the other hand if a strong base like NaOH is introduced it is buffered by H_2CO_3 and gives out NaHCO_3 (which is a weak base unlike NaOH) and H_2O
- rxn: $\text{NaOH} + \text{H}_2\text{CO}_3 \longrightarrow \text{NaHCO}_3 + \text{H}_2\text{O}$
- Henderson-Hasselbalch Equation: from this equation we can deduce that:
 - increase in HCO_3^- increases pH (shift the acid-base balance to toward alkalosis)
 - Increase in PCO_2 decreases pH (shift to acidosis) (for more info on this refer BRS page 170)

2) The phosphate buffer system

- Although not that much important as an extracellular fluid buffer, it plays a major role in buffering renal tubular fluid and intracellular fluids.
- The main elements are: H_2PO_4^- and HPO_4^{2-}
- If HCl is added to the mixture of these two elements the H is accepted by HPO_4^{2-} and become replaced by H_2PO_4^- , which is less acidic. So the decrease in pH is minimized.
- When a strong base, such as NaOH, is added to the buffer system, the OH^- is buffered by the H_2PO_4^- to form additional amounts of HPO_4^{2-} (so NaOH is replaced by NaH_2PO_4 , causing only a slight increase in pH).
- Here is the overall reaction:
 $\text{HCl} + \text{Na}_2\text{HPO}_4 \rightleftharpoons \text{NaH}_2\text{PO}_4 + \text{NaCl}$
 $\text{NaOH} + \text{NaH}_2\text{PO}_4 \rightleftharpoons \text{Na}_2\text{HPO}_4 + \text{H}_2\text{O}$
- The phosphate buffer system has a pK of 6.8, which is not far from the normal pH of 7.4 in the body fluids.

3) The Protein buffer system

- The most plentiful buffer.
- 60 to 70% of the total chemical buffering of the body fluids is in ICF
- most of this results from the intracellular proteins
- For e.g. Hb is an important buffer system in RBC



- The PK of this system is fairly close to 7.4 which is a contributing factor for their large part in the chemical buffering of the body besides their high concentration in cells.

(Pathology part)

- **Glomerulonephritis**

- Can be either primary or secondary GN.
- **Primary glomerulonephritis:** kidney itself is the principal organ involved.
- **Secondary glomerulonephritis:** Kidney is damaged by a systemic disease (e.g., hypertension, diabetes...). The primary pathology is elsewhere in other systems but there is secondary involvement of the kidneys.

Pathogenesis/mechanisms of glomerular injury

- Immune mechanisms underlie most forms of glomerular injury.
Include: - antibody mediated injury (the most important mechanism)
 - cell mediated injury
 - complement system activation
- Non-immune factors can also initiate GN or influence its progression.
- **Antibody mediated injury:** the main underlying mechanism in many forms of glomerulonephritis. There may be antibodies that are reacting against antigens they recognize =>formation of immune complexes =>activation of several effector mechanisms that can cause injury, e.g. activation of the complement system => glomerular injury. The immune complex formation can either be insitu (within the structures of the kidney), or it can be formed elsewhere then deposited within the kidneys. In insitu immune complex formation, the antigen may be intrinsic to the kidney, i.e. it may be part of the structure of the kidney, or it may be an antigen that has been planted from the circulation from elsewhere in the body. Again, the planted antigen may be endogenous in origin (from the individual's tissues elsewhere) or exogenous (microbial antigens, drugs, or toxins). Circulating immune complex deposition in the kidneys is one feature of type three hypersensitivity reactions. And this deposition can lead to glomerular injury.

Changes in glomerulus expected during GN

- Hypercellularity
- Basement membrane thickening
- Hyalinization and sclerosis (accumulation of fibrous tissue (fibrin))

Patterns of glomerular injury

- Focal glomerulonephritis: only some glomeruli involved in the disease process.
- Diffuse glomerulonephritis: most of the glomeruli in the kidney involved.
- Global process: the whole glomerulus abnormal.
- Segmental process: part of the glomerulus abnormal and part of it normal.

Location of immune complexes

- **Subendothelial:** between the endothelium and the glomerular basement membrane. The deposition of the immune complexes is in a granular fashion.
- **Glomerular basement membrane:** deposition is in a linear fashion. As the immune reaction is induced against a component of type 4 collagen which is found uniformly throughout the basement membrane, the antibodies will also come and interact with the antigens in such a way, thus resulting in a linear arrangement.

- **Subepithelial:** between the GBM and the foot processes of podocytes. Deposition is in a granular fashion.
- Immunofluorescence technique is used to identify the type of Ig that has been deposited and electron microscopy is used to define the above locations.

Clinical syndromes/presentations of GN

Acute Nephritic Syndrome

- The primary manifestation is hematuria. Also presents with azotemia, oliguria, hypertension, proteinuria, and edema. The proteinuria and edema are not as severe as that seen in nephrotic syndrome. May also present with non-specific symptoms like malaise, fever, nausea.
- The characteristic morphological pattern of glomerular injury seen in this syndrome is **diffuse proliferative GN**. Most of the glomeruli are involved + the cellularity is increased. Capillary lumina within the glomeruli appear obstructed due to the increased cellularity. The typical cause of diffuse proliferative GN is post streptococcal GN (there are also other causes).
- The hematuria can be seen as red cell casts in the renal tubules.
- **Poststreptococcal GN:** occurs 1 – 4 weeks following a streptococcal pharyngitis or streptococcal skin infection (impetigo). By the time the patient comes with the diffuse proliferative GN, the primary streptococcal infection may not be seen. All strains of group A beta-hemolytic streptococci are not associated with this complication. Only the nephritogenic strains are involved. As streptococcal infections are common in children, poststreptococcal GN is also usually a disease of young children. There will be elevated titers of anti-streptococcal antigens. The inciting antigen is either a planted or circulating streptococcal antigen. It is not very well known. Generally, poststreptococcal GN has a good prognosis. According to an EM finding, it presents with subepithelial immune complex deposition (granular pattern). The immune complexes are composed of antibodies (specifically IgG), complement proteins and whatever antigen is derived from the bacteria.
- In addition to streptococcal infections, other infections can also lead to diffuse proliferative GN:
 - Bacterial infections
 - Staphylococcal endocarditis
 - Pneumococcal pneumonia
 - Meningococcemia
 - Parasitic infections
 - Malaria
 - Toxoplasmosis
 - Viral infections
 - HBV, HCV - such infections are usually chronic infections. The antigen will be continuously released into the blood stream. Thus infection with these viruses and the GN can simultaneously take place. Therefore, not strictly postinfectious GN.

HIV, Mumps, Varicella, Infectious mononucleosis

-Endogenous

SLE – an autoimmune disease. The prognosis is poor, because there is a continuous source of antigens => there'll be chronic inflammation=> a higher chance of progression to chronic renal failure.

Rapidly progressive glomerulonephritis

- Characterized clinically by a rapid, progressive renal decline.
- Its characteristic morphological pattern is **crescentic GN**.
- Its difference from nephritic syndrome is that it shows development into renal failure.
- Can either have primary (idiopathic) or secondary forms.
- Crescentic GN is made up of inflammatory cells, endogenous glomerular epithelial cells, and fibrin. There is a rupture in the GBM (under EM). There is leakage of higher molecular weight proteins including fibrinogen. Generally, has a poor prognosis.
- It has three groups:
- **Type I RPGN**: is an anti-GBM disease characterized by linear IgG and C3 (complement component) GBM deposits. In some cases, the anti-GBM antibodies cross-react with pulmonary alveolar basement membranes to produce pulmonary hemorrhages – this condition where there is involvement of both kidneys and lungs is called **Goodpasture syndrome**. What stimulates the antibody production is not known. The antibodies react against components of type 4 collagen found in basement membranes. It manifests in the lungs as cough, shortness of breath, and hemoptysis.
- **Type II RPGN**: is an immune complex-mediated disease. Can present as a complication of any of the immune complex nephritides, e.g. postinfectious GN and SLE. Shows characteristic granular staining.
- **Type III RPGN**: presents with vascular abnormalities that give crescent formation. Also known as pauci-immune, because there is no immune complex deposition as in the other types. It is associated with systemic vasculitis syndromes – e.g. Wegner's syndrome and polyarthritis nodosa (PAN).

Nephrotic syndrome

- Its primary manifestation is proteinuria. In order to be called nephrotic the proteinuria should be $\geq 3.5\text{g/day}$.
- Also presents with hypoalbuminemia, generalized edema, hyperlipidemia, and lipiduria.
- Patients are more susceptible to certain infections, especially staphylococcal and pneumococcal infections. This is because, due to the proteinuria there is loss of globulin proteins along with other proteins through the urine, and it is the antibodies (Ig s) that are responsible for defense against such pyogenic bacteria (and they are made with proteins like globulins).

- Patients might also lose components of the coagulation/anticoagulation cascade. This can predispose to thrombotic and thromboembolic events - Renal vein thrombosis is a common consequence of nephrotic syndrome.
- In few patients => hematuria and hypertension.
- There are systemic diseases that have nephritic syndrome as one manifestation:
 1. Diabetes mellitus
 2. Amyloidosis
 3. SLE
 4. Drugs
 5. Infections
 6. Malignancies
- In many cases of systemic causes of nephrotic syndrome the pattern that is seen is the membranous pattern of glomerulopathy (discussed below).
- Nephrotic syndrome, unlike nephritic syndrome, has many patterns of glomerular abnormalities (primary glomerular diseases):

1. **Membranous glomerulopathy:**

- Can be idiopathic or secondary. When it is secondary it commonly results from DM, amyloidosis, malignancies etc.
- Membranous glomerulopathy is the most common cause of nephrotic syndrome in adults. Whereas the minimal change disease (discussed below) is the most common cause of nephrotic syndrome in children.
- Similar to nephritic syndrome, there is activation of complement by the classical pathway. There is also immune complex deposition in the glomeruli (it is immune-complex mediated). The immune complex is deposited in the subepithelial location (thus has a granular pattern of deposition).
- Membranous glomerulopathy involves effacement of foot processes. Effacement of foot processes is seen in many of the glomerular abnormalities which result in nephrotic syndrome. Whenever there is glomerular disease that presents with marked proteinuria, there will be involvement of the foot processes – as they are important in the selective permeability of glomerular filtration barrier.
- It's called membranous glomerulopathy because under light microscopy the glomerular BM appears thicker than normal. The BM becomes more prominent. The BM appears as such because there is deposition of immune complexes.
- The prognosis is variable. It is not as uniformly excellent as in post streptococcal GN. Proteinuria persists in >60% cases. 10% progress to renal failure within 10 years. Eventually up to 40% cases develop renal insufficiency.

2. **Minimal change disease**

- Most common cause of nephrotic syndrome in children. It is called minimal change because under light microscope the glomeruli appear normal - cannot be differentiated from normal glomeruli. The only abnormality presented is effacement

of the foot processes. Unlike membranous glomerulopathy, it does not involve immune complex deposition.

- This disease has features that suggest its immunologic basis:
 - Association with immunization & URTIs (upper respiratory tract infections)
 - Response to immunosuppressive therapy - highly responsive to corticosteroid therapy
 - Association with atopic disorders - which are immunologic based diseases
 - Has some genetic predispositions
- Has good prognosis - majority of the cases will have complete resolution of the nephrotic syndrome.

3. Focal segmental glomerulosclerosis

- Can be a manifestation of more severe forms of other glomerulonephritis (complication of other GN). For example, more severe form of IgA nephropathy (one pattern of GN seen in nephrotic syndrome-discussed below) can lead to FSGS.
- FSGS can be a congenital defect. Patients with mutation of their podocyte molecules/proteins, e.g. like nephrin and podocin, can develop nephrotic syndrome during infancy.
- There can also be secondary forms of FSGS, e.g. **HIV nephropathy**.
- Involves effacement of foot processes like the membranous glomerulopathy and minimal change disease.
- Initially, it affects only the juxtamedullary glomeruli. But as it becomes advanced, more glomeruli will be involved and the process would be global instead of segmental – more generalized, global sclerosis will be presented in the later stages.
- Clinically patients with FSGS have a higher incidence of mixed picture of both nephritic and nephrotic syndromes (i.e. In addition to presenting with the nephrotic syndrome, FSGS patients also show hematuria and hypertension, which are features of nephritic syndrome).
- Has poorer prognosis - at least 50% cases develop end-stage renal failure within 10 years of diagnosis.

4. Membranoproliferative GN

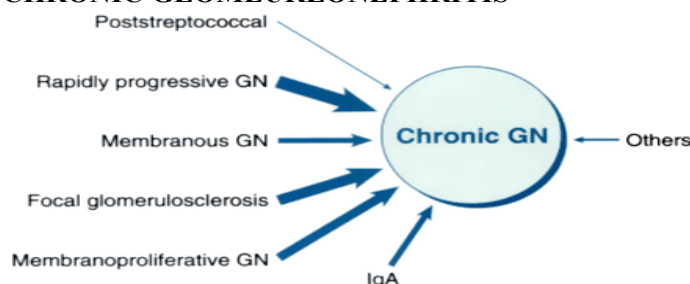
- It can be idiopathic or secondary to another disorder or agent.
- It is categorized into two forms:
 - **Type I** – is the most common form. It is likely a consequence of antigen-antibody complex deposition and complement activation. The antigen can originate from infection (e.g., hepatitis B or C, endocarditis, HIV), SLE, or malignancy, but in most cases the source is unknown (idiopathic).

- **Type II** – called dense-deposit disease. Occurs due to activation of the alternate complement pathway.
- Under light microscopy, glomeruli appear “lobular” due to mesangial proliferation.
- Under EM, the type I MPGN shows subendothelial electron-dense deposits, whereas the type II MPGN shows Characteristic laminar, ribbon-like electron-dense deposits in the GBM proper.
- Most of the patients are at a young age group - adolescence or young adulthood. They present with nephrotic syndrome, occasionally with hematuria.
- Has poor prognosis - about 50% of patients develop chronic renal failure within 10 years. There is a high recurrence rate in transplant recipients, particularly in patients with type II disease.

5. IgA nephropathy (Berger’s disease)

- Is probably the most common type of glomerulonephritis worldwide and is a major cause of recurrent glomerular hematuria.
- The classical presentation for IgA nephropathy is presentation with gross hematuria.
- It can manifest with almost any of the glomerular syndromes.
- It can either be primary or secondary. When primary, the antigen that elicits antibody response is unknown (idiopathic). In secondary form, for e.g. it can occur due to liver diseases. Patients with chronic liver disease have defect in clearing the IgA that is synthesized and normally present in the serum. Because of this there will be excess amount of IgA. Increased systemic IgA or qualitative alterations in the IgA molecule leads to augmented deposition in the mesangium and subsequent activation of the alternate complement pathway. This can lead to development of secondary IgA nephropathy. This represents only the minority of IgA nephropathy patients.
- The majority of the cases are primary IgA nephropathy cases.
- It is more common in younger children.
- The typical site within the glomerulus that is involved in IgA nephropathy is the mesangium – there is a mesangial immune complex deposition.
- IgA nephropathy shows some genetic predisposition.
- The Ig deposited is obviously IgA. This is unlike most of the other GN, where the most common Ig deposited is IgG. Immunofluorescence is important to detect the IgA (important for diagnosis of IgA nephropathy).
- Under LM, it can show a normal looking glomerulus or range of microscopic abnormalities - mesangioproliferative GN, focal proliferative GN, crescentic GN, or focal segmental G sclerosis.
- Most of the time IgA nephropathy develops following URT infections, but there is no specific infectious agent associated with it.
- Prognosis - slow progression to CRF in ~25%. Extent of glomerulosclerosis shows correlation with prognosis.

CHRONIC GLOMERULONEPHRITIS



-Late stage of several types of glomerular diseases. (Fig. thickness of the arrows indicates the likelihood of progressing into chronic GN)

- May also develop in the absence of a recognized acute GN, i.e. the acute GN may have been undiagnosed or may have been asymptomatic.
- Chronic GN => Chronic RF (clinically).
- Clinically, it may present with non-specific symptoms like loss of appetite, nausea and weakness- these may be manifestations of uremia. Proteinuria is presented in relatively earlier stages of chronic GN, it subsides in advanced stages. It also presents with hypertension, which is a picture of the RF (RF=> failure of the glomerular filtration=> retention of waste products, fluids and salts=> hypertension).
- Once a patient develops chronic GN, through time, he/she show progression to end stage renal disease. This is the most advanced stage of RF. At end stage renal disease, the patient needs either dialysis or transplantation. Thus, once a patient gets to the stage of chronic GN the process is not reversible.
- Morphologically, it presents with shrunken, symmetrically contracted kidneys. In addition, there would be cortical atrophy – the cortex appears thinner in cross section of the kidneys. There might be involvement of the tubules and the interstitium – they might be secondarily affected. There could also be vascular abnormalities in chronic GN. In most cases this occurs due to the hypertension.
- Thus, chronic GN = abnormal glomeruli + abnormal tubules/ interstitium + abnormal vascular compartment.
- There is also systemic involvement in chronic GN=> uremic pericarditis, uremic gastroenteritis, nephrocalcinosis (accumulation of calcium in renal tissue), left ventricular hypertrophy (due to the hypertension), and diffuse alveolar damage.

• **Tubulointerstitial Nephritis**

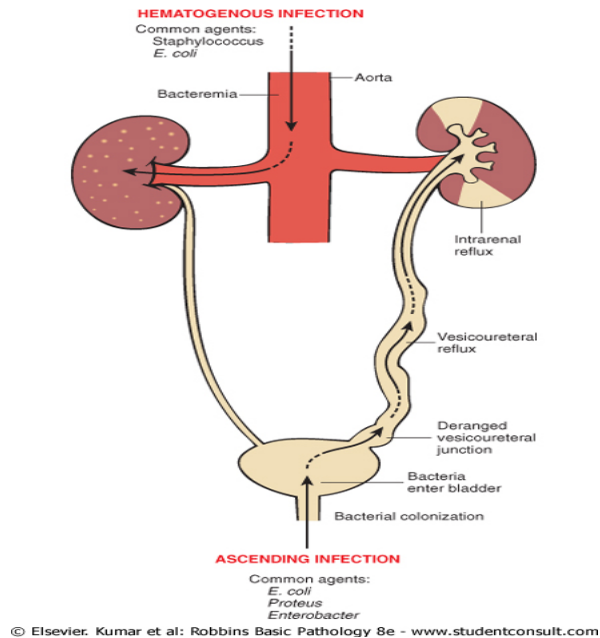
- Is a group of inflammatory diseases of the kidneys that primarily involve the interstitium and tubules.
- Is among the two major types of tubulointerstitial diseases.
- Tubulointerstitial diseases =>
 1. Tubulointerstitial nephritis – primarily inflammatory
 2. Acute tubular necrosis - primarily manifested as injury at the cellular level.
- TIN has diverse causes and pathogenic mechanisms:
 - ◆ **Infections** - pyelonephritis
 - ◆ Toxins
 - ◆ Metabolic diseases
 - ◆ Neoplasms
 - ◆ Physical factors
 - ◆ Immunologic disorders
- TIN are distinguished from glomerular diseases by the absence of the hallmarks of nephritic or nephrotic syndrome and by the features of tubular dysfunction (e.g., impaired urine-concentrating ability [polyuria], salt wasting, and metabolic acidosis).

Pyelonephritis

- Infection of the renal interstitium and tubules with prominent involvement of the renal pelvis. It mostly involves bacterial infections, but there could also be viral and fungal pyelonephritis.
- Among the causes of TIN, bacterial infection and analgesic nephropathy present with prominent involvement of the renal pelvis.
- Pyelonephritis can be acute or chronic.

Acute Pyelonephritis:

- Suppurative inflammation of the kidney and the renal pelvis caused by bacterial infection. The tubules, interstitium, and renal pelvis are involved. It is considered as upper urinary tract infection (UTI).
- Pathogenesis - Mostly acute pyelonephritis is an endogenous infection. The bacteria are derived from the normal flora of the patient – meaning normal flora of the GIT. Thus, if bacteria that are part of the normal flora of the GIT gain access to the UT then they can cause UTI. E. coli is the most common bacterial cause. Proteus, Klebsiella, Enterobacter, and Pseudomonas can also be causes (enteric G- negative rods).
- There are two different routes of infection that can lead to acute pyelonephritis. These are hematogenous and ascending routes. The ascending route of infection is the commoner one. Ascending route of infection occurs through stages. It starts with colonization of urethra and bladder. In this colonization bacterial virulence factors like adhesions and fimbriae are involved. The flushing action of urine is one of the defense mechanisms against colonization. If there is urine stasis then this defense mechanism would be lost. So, once the bladder is colonized, there will be lower UTI. Lower UTI is a very common type of infection. Its symptoms include – burning urine, increased frequency of urine, irritation etc. Among patients who have lower UTI, some develop upper UTI. Lower UTI may not always be symptomatic. Sometimes patients may develop acute pyelonephritis with no previous symptoms of lower UTI.
- The main predisposing condition for developing upper UTI (acute pyelonephritis) is the presence of vesicoureteral reflux – a condition in which there is weakened compression of the lower ends of the ureters or in which the intravesical portions of the ureters are short, and because of this there is reflux of urine back into the distal portions of the ureter. Under normal circumstances, during micturition, the bladder muscle contracts and there will be compression of the lower ends of the ureter so there would not be a reflux back into the ureter. Vesicoureteral reflux can be congenital or acquired. Congenital vesicoureteral reflux usually manifests as pyelonephritis in young children, whereas the acquired one is common among adults.



- Once the urine is refluxed back into the renal pelvis then there will be intrarenal reflux. Then the urine refluxes back thru the renal pelvis, thru the papillae into the tubules. This can be a precondition for developing acute pyelonephritis.
- Predisposing conditions:
 - ◆ Urinary tract obstruction – leads to stasis or reflux
 - ◆ Instrumentation
 - ◆ Vesicoureteral reflux
 - ◆ Pregnancy – Bc there is smooth muscle relaxation and compression
 - ◆ Pre-existing renal lesions
 - ◆ Diabetes mellitus - Bc of immunocompromisation and autonomic nervous system dysfunction
 - ◆ Immunodeficiency
- Uncomplicated pyelonephritis follows a benign course with antibiotic therapy but can recur or progress in the presence of vesicoureteral reflux, obstruction, immunocompromise, diabetes, and other conditions. Generally, acute pyelonephritis clinically presents with costovertebral (flank) angle pain, fever, malaise, abnormal urine sediment, pyuria, WBC casts, and symptoms of lower UTI (Bc mostly there is ascending route of infection)
- Morphology - patchy interstitial suppurative inflammation, intratubular aggregates of neutrophils, and tubular necrosis.
- Papillary necrosis is a dangerous complication of acute pyelonephritis. It is usually a bilateral process. If all the renal papilla are necrotized then the kidney will be unable

to function. Then it can lead to renal failure. Patients with diabetes are at high risk of developing this complication from acute pyelonephritis.

Chronic Pyelonephritis

- It is a chronic tubulo-interstitial inflammation caused by infection.
 - Differs from the acute pyelonephritis in that there is fibrosis and scarring within the interstitium and the pelvicalyceal region.
 - Has two forms:
 - ♦ **Reflux nephropathy** – is the most common. It begins in childhood, as a result of infections superimposed on congenital vesicoureteral reflux and intrarenal reflux. It can be unilateral or bilateral.
 - ♦ **Chronic obstructive pyelonephritis** - occurs when chronic obstruction (e.g., with hydronephrosis) predisposes the kidney to infections. The effects of chronic obstruction also contribute to parenchymal atrophy.
 - Chronic pyelonephritis, like chronic GN, is an important cause of chronic renal failure.
 - Both major types of chronic pyelonephritis are associated with broad scars, deformed calyces, and significant tubulointerstitial inflammation and fibrosis. Secondary focal segmental glomerulosclerosis (FSGS) (due to loss of glomerular mass) and vascular hypertensive changes can also be present. In chronic pyelonephritis, the gross abnormalities seen in the two kidneys is asymmetrical/irregular, this is unlike that which is seen in chronic GN, which presents with symmetrical involvement of the kidneys.
 - Clinically, both forms of CPN can manifest with the symptoms of acute pyelonephritis or can have a silent, insidious onset (Insidious onset of hypertension or chronic renal failure). The development of proteinuria and FSGS is a poor prognostic sign.
- **Tumors of the urinary tract and kidney**

Benign tumors

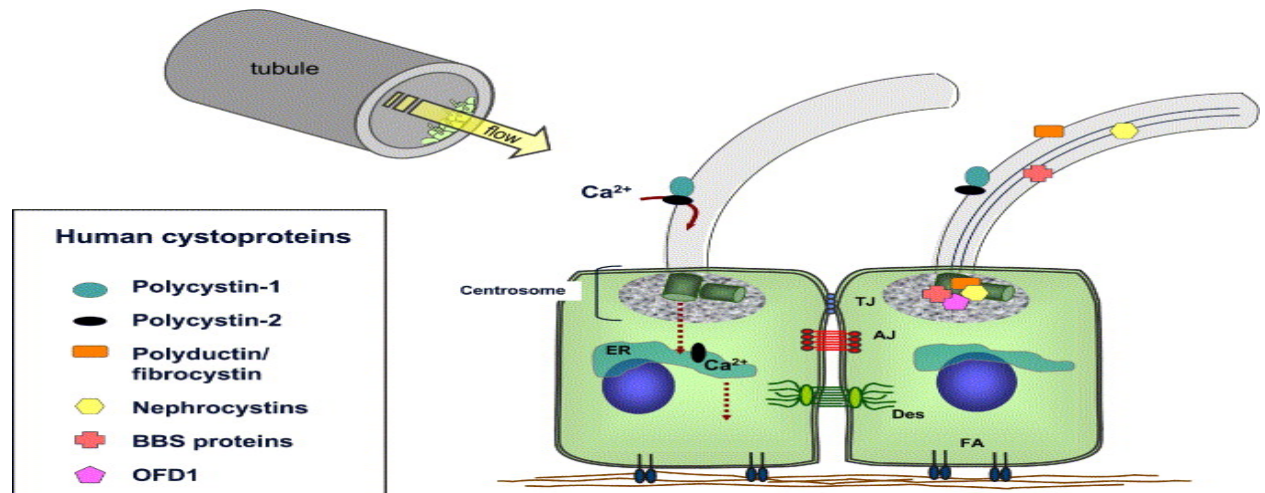
- **Oncocytoma**: is an epithelial tumor composed of eosinophilic cells arising from collecting duct intercalated cells. The cells are packed with mitochondria (under EM). These tumors are common and can be large.
- **Papillary adenomas**: are common, small, yellow cortical tumors. Involve the renal tubular epithelium. Histologically, most consist of vacuolated epithelial cells forming tubules and complex branching papillary structures. They are histologically indistinguishable from low-grade papillary renal cell carcinomas, and they share some of their cytogenetic features.

Malignant tumors

- **Renal cell carcinoma (Adenocarcinoma):** usually occur in patients between the ages of 50 and 70 years and show a 2:1 male preponderance. Tobacco is the most significant risk factor. Obesity, hypertension, unopposed estrogens, and exposures to asbestos, petroleum products, and heavy metals are also implicated. Most cases are sporadic, but autosomal dominant familial variants account for 4% of cases.
- RCC can be classified based on histology, cytogenetics, and genetics:
 - **Clear cell carcinoma:** is the most common type. Most of the cases are sporadic. However, in 98% of the cases, there is a loss of sequences on chromosome 3p at a locus that harbors VHL (a tumor suppressor gene that encodes part of a ubiquitin ligase complex involved in targeting proteins for degradation). Thus, the cause for this tumor can also be genetic – mutation of VHL.
 - **Papillary carcinoma:** occurs in both familial and sporadic forms. The familial form is associated with mutations of the MET proto-oncogene that serves as a tyrosine kinase receptor for hepatocyte growth factor. These tumors can be multifocal and bilateral. It is the most common type in dialysis-associated cases.
 - **Chromophobe renal cell carcinoma:** derive from collecting duct intercalated cells. Present with large pale cells with prominent cell membranes. Although they exhibit multiple chromosome losses and extreme hypodiploidy, they have an excellent prognosis.
- Patients present with flank pain, palpable mass, and hematuria. Renal cell carcinomas also produce a host of paraneoplastic syndromes attributable to hormone production:
 - ◆ Polycythemia
 - ◆ Hypercalcemia
 - ◆ Hypertension
 - ◆ Feminization/masculinization
 - ◆ Cushing syndrome
 - ◆ Eosinophilia
 - ◆ leukemoid reaction
 - ◆ amyloidosis
 - ◆ Pseudohyperparathyroidism
- Prognosis depends on tumor size and the extent of spread at diagnosis. Renal cell carcinoma tends to metastasize before symptoms are felt.
- **Nephroblastoma (Wilm's Tumor):**
 - ◆ the most common primary renal tumor in children
 - ◆ may arise sporadically or may be familial - with the susceptibility to tumorigenesis
 - ◆ Inherited as an autosomal dominant trait.
 - ◆ Malignant embryonal neoplasm
 - ◆ Derived from nephrogenic blastemal cells

- ◆ Triphasic-blastemal, stromal, and epithelial cell types
 - ◆ Clinically presents with abdominal mass, hematuria, abdominal pain, intestinal obstruction, and hypertension.
 - ◆ Good prognosis (85% cured)
- **Urothelial Neoplasms:**
 - ◆ Account for 5-10% of primary renal tumors
 - ◆ Common site - bladder
 - ◆ Most common risk factors - analgesic abuse
 - ◆ Spectrum – benign to malignant
 - ◆ Prognosis – depends on grade and stage
- **Cystic Diseases of the Kidney**
 - Are a heterogeneous group comprising hereditary, developmental, and acquired disorders. They are non-neoplastic lesions. They include:
 - **Cystic Renal Dysplasia:**
 - ◆ Abnormal metanephric differentiation
 - ◆ Associated ureteropelvic obstruction, ureteric agenesis, atresia or reflux
 - ◆ Immature mesenchyme
 - ◆ Immature collecting ducts
 - **Simple Cysts:** are Single or multiple cysts of the cortex and rarely the medulla. They are lined by a single layer of low cuboidal epithelium. They are common lesions. They have smooth walls and are filled with clear serous fluid. The main importance of these cysts lies in their differentiation from kidney tumors (i.e., they mimic renal carcinomas). However, radiographic studies show that in contrast with renal tumors, renal cysts have smooth contours, are almost always avascular, and produce fluid rather than solid tissue signals on ultrasonography.
 - **Adult Polycystic Kidney Disease:** aka autosomal dominant polycystic kidney disease. It accounts for 5% to 10% of chronic renal failure. It has high penetrance and is universally bilateral.
 - ADPKD is caused in most cases by mutations in one of two genes:
 - ◆ **PKD1 gene mutations** – account for most of the cases. PKD1 encodes polycystin 1, a large protein that localizes to tubular epithelial cells and has domains that are usually involved in cell-cell and cell-matrix interactions.
 - ◆ **PKD2 gene mutations** - encodes polycystin2, a cation channel. These mutations disrupt the regulation of intracellular calcium.
 - Pathogenesis - revolves around the sensing and transduction of mechanical signals. A single apical non-motile primary cilium in tubular epithelial cells functions as a mechanosensor to monitor changes in fluid flow and shear stress, while intercellular junctional complexes and focal adhesions monitor forces between cells and extracellular matrix. In response to external forces, these sensors regulate ion flux that, in turn, modulates cell polarity and proliferation. Polycystin 1 and 2 are both localized

to the primary cilium and potentially form a complex that regulates intracellular calcium in response to fluid flow. Mutated proteins could conceivably affect intracellular second messengers and thereby influence proliferation, apoptosis, ECM interactions, and secretory function, leading to the progressive formation of tubular cysts.



- Kidneys are massively enlarged and composed almost entirely of cysts. Cysts arise anywhere along the nephron and compress adjacent parenchyma. In late disease, there is interstitial inflammation and fibrosis.
- Normal renal function is maintained into adulthood (until the renal insufficiency supervenes). Cystic dilation or hemorrhage can cause pain, hematuria, hypertension, proteinuria, and polyuria. Progression is accentuated in the presence of hypertension. Patients with this disease may also present with other extrarenal congenital anomalies: => Polycystic liver disease, Berry aneurysms, mitral valve prolapse, and Splenic, pancreatic, or pulmonary cysts.
- **Childhood polycystic kidney disease:** aka autosomal recessive PKD. It is genetically distinct from ADPKD. In most cases, the disease is caused by mutations of PKHD1, encoding for fibrocystin, a large transmembrane protein that localizes to the primary cilium of tubular epithelial cells. Kidneys are enlarged by multiple, cylindrically dilated collecting ducts oriented at right angles to the cortex and filling both the cortex and medulla. The liver almost always has cysts and proliferating bile ducts.
- Can be categorized by age of presentation as => Perinatal, neonatal, infantile, and juvenile.

• Urinary tract obstruction

Renal stones

- Urolithiasis is calculus formation at any level in the urinary collecting system, but most often the calculi arise in the kidney.

- They occur frequently.
- Symptomatic urolithiasis is more common in men than in women.
- Stones are unilateral in about 80% of patients.
- Common sites of formation are renal pelvis and calyces and the bladder.
- A confluence of predisposing conditions, including the concentration of the solute, changes in urine pH, and bacterial infections.
- The most important cause is increased urinary concentration of the stone's constituents, so that it exceeds their solubility in urine (supersaturation).

Pathogenesis

- There are three major types of stones.

1-Calcium oxalate and/or calcium phosphate.

- Most common type and usually seen in adults.
- Have hypercalciuria that is not associated with hypercalcemia.
- Most in this group absorb calcium from the gut in excessive amounts (absorptive hypercalciuria) and promptly excrete it in the urine, and some have a primary renal defect of calcium reabsorption (renal hypercalciuria).

2- Magnesium ammonium phosphate (struvite) stones

- Second most common type
- Almost always occur in persons with a persistently alkaline urine resulting from UTIs. In particular infections with urea-splitting bacteria, such as *Proteus vulgaris* and staphylococci.
- Classically results in staghorn calculi (branching structures which create a cast of the renal pelvis and calyceal system in renal calyces. (Cause of staghorn calculi in adults)

3- Uric acid or cysteine stones

- **Uric acid stone:**
- Third most common type.
- Risk factors include hot, arid climate, low urine volume and acidic PH.
- Gout and diseases involving rapid cell turnover, such as the leukemia, lead to high uric acid levels in the urine and the possibility of uric acid stones.
- it is radiolucent (can't detect in x-ray) as opposed to other types of stones which are radiopaque.

- **Cysteine stone**
- very rare
- Most commonly seen in children.
- Associated with cystinuria (a genetic defect on tubules that result in decreased reabsorption of cysteine).
- Like uric acid stones, cysteine stones are more likely to form when the urine is relatively acidic.
- Cause of staghorn calculi in children.

Clinical course

- Stones may be present without producing either symptoms or significant renal damage. This is particularly true with large stones lodged in the renal pelvis.
- Smaller stones may pass into the ureter, where they may lodge, producing a typical intense pain known as renal or ureteral colic, characterized by paroxysms of flank pain radiating toward the groin. Often at this time there is gross hematuria.
- The clinical significance of stones lies in their capacity to obstruct urine flow or to produce sufficient trauma to cause ulceration and bleeding. In either case, they predispose the sufferer to bacterial infection.
- Fortunately, in most cases the diagnosis is readily made radiologically.

Hydronephrosis

- Hydronephrosis refers to dilation of the renal pelvis and calyces, with accompanying atrophy of the parenchyma, caused by obstruction to the outflow of urine.
- The obstruction may be sudden or insidious, and it may occur at any level of the urinary tract, from the urethra to the renal pelvis.
- The most common causes are categorized as follows:
- **Congenital:** atresia of the urethra, valve formations in either ureter or urethra, aberrant renal artery compressing the ureter, renal ptosis with torsion, or kinking of the ureter.
- **Acquired**
 - Foreign bodies: calculi, sloughed necrotic papillae

- Proliferative lesions: benign prostatic hyperplasia, carcinoma of the prostate, bladder tumors (papilloma and carcinoma), contiguous malignant disease (retroperitoneal lymphoma, carcinoma of the cervix or uterus)
- Inflammation: prostatitis, ureteritis, retroperitoneal fibrosis.
- Neurogenic: spinal cord damage with paralysis of the bladder
- Normal pregnancy: mild and reversible
- Bilateral hydronephrosis occurs only when the obstruction is below the level of the ureters.
- Sometimes obstruction is complete, allowing no urine to pass; usually it is only partial.

Pathogenesis

- Even with complete obstruction, glomerular filtration persists for some time, and the filtrate subsequently diffuses back into the renal interstitium and perirenal spaces, whence it ultimately returns to the lymphatic and venous systems.
- Because of the continued filtration, the affected calyces and pelvis become dilated.
- The unusually high pressure thus generated in the renal pelvis, as well as that transmitted back through the collecting ducts, causes compression of the renal vasculature. Both arterial insufficiency and venous stasis result, although the latter probably is more important.
- The most severe effects are seen in the papillae, because they are subjected to the greatest increases in pressure.
- Accordingly, the initial functional disturbances are largely tubular, manifested primarily by impaired concentrating ability. Only later does glomerular filtration begin to diminish.
- Serious irreversible damage occurs in about 3 weeks with complete obstruction, and in 3 months with incomplete obstruction.
- In addition to functional changes, the obstruction also triggers an interstitial inflammatory reaction, leading eventually to interstitial fibrosis.

Morphology

- With subtotal or intermittent obstruction, the kidney may be massively enlarged. The renal parenchyma itself is compressed and atrophied, with obliteration of the papillae and flattening of the pyramids.

- On the other hand, when obstruction is sudden and complete, glomerular filtration is compromised relatively early, and as a consequence, renal function may cease while dilation is still comparatively slight.
- Depending on the level of the obstruction, one or both ureters may be dilated (hydroureter).
- On microscopic examination the early lesions show tubular dilation, followed by atrophy and fibrous replacement of the tubular epithelium with relative sparing of the glomeruli.
- Eventually, in severe cases the glomeruli also become atrophic and disappear, converting the entire kidney into a thin shell of fibrous tissue.
- With sudden and complete obstruction, there may be coagulative necrosis of the renal papillae, similar to the changes of papillary necrosis.

Clinical course

- Bilateral complete obstruction produces anuria, which is soon brought to medical attention.
- When the obstruction is below the bladder, the dominant symptoms are those of bladder distention.
- Paradoxically, incomplete bilateral obstruction causes polyuria rather than oliguria, as a result of defects in tubular concentrating mechanisms.
- Unfortunately, unilateral hydronephrosis may remain completely silent for long periods unless the other kidney is for some reason not functioning.
- Removal of obstruction within a few weeks usually permits full return of function; however, with time the changes become irreversible.

(Toxicology part)

• **Routs of poisoning and types of toxicity**

- Agents generally produce the greatest effect and the most rapid response when given directly into the bloodstream (the intravenous route).
- An approximate descending order of effectiveness for the other routes would be inhalation, intraperitoneal, subcutaneous, intramuscular, intradermal, oral, and dermal.

- In addition, the route of administration can influence the toxicity of agents. For example, an agent that acts on the CNS, but is efficiently detoxified in the liver, would be expected to be less toxic when given orally than when given via Inhalation.
- The major routes by which toxic agents gain access to the body are through the gastrointestinal tract (ingestion), the lungs (inhalation), or the skin (topical, percutaneous, or dermal).

Oral

- Most common route of exposure
- Responsible for all causes of poisoning
- Suicidal poisoning occurs most frequently by oral ingestion

Inhalational

- Fumes, particulate matters and gases
- Particles < 1µm penetrate lower airways
- Lipid soluble agents have access to be absorbed
- Water soluble toxicants irritate respiratory tract
- Removal of victim from toxic atmosphere reduces absorption
- Responsible for accidental, therapeutic and occupational poisoning

Cutaneous

- Lipid soluble agents are highly absorbed
- Inflammation/rubbing increase absorption (because of vasodilation)
- Ice/torniquets retard absorption (because of vasoconstriction)
- Responsible for accidental, therapeutic and occupational poisoning

- **Diagnosis of poisoning**

- History, Physical examination, Lab. Investigation and Toxicological analysis

- **History:** what substance? At what time? In what amount?

- **Physical examination:**

Vital signs (heart rate, blood pressure, respiratory rate, temperature). System based assessment (looking in to the eyes, GIT assessment, circulatory system assessment...).

Toxidromes (signs and symptoms an intoxicated person shows at simultaneous time and usually specific to a certain group of toxicants or even a particular toxicant) are also important. Example of toxidromes => cholinergic drugs show miosis, bradycardia, excessive salivation, sweating, hyperactivity of bowel (urination, defecation).

- **Lab. Investigation:**

An anionic gap (if high it indicates metabolic acidosis that can be caused by alcohols, cyanides, salicylates...).

Anion gap = $[Na^+] - ([Cl^-] + [HCO_3^-])$;

Osmolality gap and organ functioning tests although not done as common as anionic gap they are important.

Osmolality gap = measured osmolality – calculated osmolality

- **Toxicological analysis:**

It is done for a late evaluation and prediction of quantitative and qualitative analysis. The first step is to select an appropriate specimen (usually blood and urine but also vomitus, autopsy, biopsy...). Then do chemical test (qualitative) and/or chromatography (quantitative).

- The chromatography (a process in which a chemical mixture carried by a liquid or gas is separated into components as a result of differential distribution of the solutes as they flow around or over a stationary liquid or solid phase) can be gas chromatography or high pressure liquid chromatography. Another quantitative assay is the immune assay.

- **Management of poisoning**

- The four fundamental measures in managing poison:

- A. **Supportive care** – it is the most important. Its objective is to keep the person alive and maintain homeostasis until we undergo detoxification process to avoid or reduce secondary complications. Example – in case of airway obstruction we need to clear the airway, oxygen supplement in case of respiratory failure, IV saline/vasopressor in case of circulatory shock,

controlling body temperature and managing metabolic disorder. Sometimes need to intubate (insert a tube in the trachea) the patient, primarily to avoid gastric aspirate.

- B. **Decrease further absorption** – if it is due to external exposure you need to gently wash with water and soap. Vigorous washing is not recommended as it can increase absorption. If it is taken through the GIT, we can use certain four techniques.

1. **Induction of emesis** – ipecac syrup is used for a chemical induction. It stimulates the chemo-trigger zone to induce emesis within 15-20 minutes. not used, if the patient isn't adequately conscious or if the poison is aromatic hydrocarbon (both increase risk for gastric aspirate which may give rise to pneumonitis), if the poison is corrosive (strong acid/base which can give rise to esophageal erosion) and if the poison can cause rapid deterioration (tricyclic antidepressant, cyanide, calcium channel blockers...) which can cause unconsciousness. It is still recommended if the patient is far away from a health institution.

2. **Gastric Lavage** – it is done by using either warm water or saline (250ml through nasogastric tube is recommended). After administering the fluid you drain it out repeatedly until you see a clear solution coming out of the stomach. The left lateral decubitus position is the best as it allows adequate entrance of the tube in to the stomach and markedly reduce the risk for gastric aspiration. We can intubate the patient if the patient is not adequately conscious or if we suspect the toxicant causes rapid deterioration.

3. **Adsorbent (activated charcoal)** – can adsorb pharmacological substances (cardiac glycosides, benzodiazepines, tricyclic antidepressants, phenobarbital, carbamazepine, salicylates). But we don't use activated charcoal for substances that doesn't have the affinity to bind to it (iron, lithium, inorganic compounds, ethanol, methanol, cyanide).

4. **Whole bowel irrigation** – with the use of osmotic laxative, polyethylene glycol or/and electrolyte solution. It is usually through nasogastric tube in a sitting position. It is implicated if multi-dose activated charcoal fails. It's contraindicated in caustic (strong base) ingestion.

- C. **Enhance elimination** – it is done once the drug has accessed the circulation. Osmotic diuresis can be done but the best option is altering the urinary PH. We alter the urinary PH in order to keep the drugs in the ionized form so that elimination is enhanced. Make the urine basic (a bicarbonate injection) for a drug which is a weak acid (like salicylates, sulfonamides, barbiturates), make the urine acidic for a drug which is a weak base. The other approach is to

use extracorporeal technique to facilitate clearance of toxins. This includes hemodialysis, hemoperfusion and others... In all the blood used is a venous blood.

1. Hemodialysis – the patient's blood will pass in an external container. This allows the movement of toxicants through the dialyzing membrane (Millipore membrane). It is used when the poisoning becomes lethal or the body can't clear it physiologically. The blood is one side of the dialyzing membrane and the dialysate solution in the other side. The dialysate solution is identical to a normal blood in its osmolality. As a result the toxin moves through the membrane escaping from the blood. Finally toxin free blood returns to the body. Hemodialysis can also correct the acid-base disturbance and metabolic abnormalities.

There are factors that affect the effectiveness of hemodialysis:

- The lower the molecular weight the more effective the procedure,
- The lower the volume of distribution the more effective the procedure,
- The lower the plasma protein binding capacity the more effective the procedure,

Intrinsic clearance also matters – it is due to a short half-life of the drug

Risks associated with hemodialysis

- Possible blood coagulation – requires anticoagulant therapy

2. Hemoperfusion – in this case there is an adsorbent filled in a cartridge. Venous blood will be pumped through this cartridge then back to the patient.

Factors that affect effectiveness of hemoperfusion

- Can the substance be adsorbed with a charcoal?
- Volume of distribution
- Intrinsic clearance
- Plasma protein binding capacity will not affect hemoperfusion as long as the drug has an affinity for the charcoal.

Risks associated with hemoperfusion

- Blood coagulation ... anticoagulant therapy needed
- Platelet can adsorb charcoal leading to thrombocytopenia
- There are also other less effective extracorporeal techniques, but they are only used in case the above 2 two are not available or are contraindicated. They are listed below:
 - Exchange transfusion/whole blood exchange – used when there is hemolysis (hemoglobinemia)
 - Peritoneal dialysis – relatively cheap and very available but less effective

- Plasma floresis – take portion of blood, filter it, discard the filtrate, and reconstitute the remaining with fresh frozen plasma, reinfusion to the patient. Has its own risk and effectiveness.
- Hemofiltration – a continuous process. The blood is filtered and sterile solution will be reconstituting.

4. Antidote – it is a substance that can counteract the toxic effect of the poison. It is an effective, non-noxious and easy/uncomplicated. Their limitation is that they may not be found or they may not be accessible at some places, even in health institutions. They can be of three types depending on their mechanism:

- Chemical based (simple interaction with the toxicant)
- Receptor based (agonist or antagonist for the receptor)
- Functional based (reverse the toxic effect directly or indirectly)

- **Potential causes and sources of poisoning**

Causes of poisoning

- Suicide (youth), Homicide (illiterates), Accident (children), Therapy (outpatients and inpatients), Occupation

SOURCES OF TOXIC COMPOUNDS

- The potential causes of toxicities generally include:
 - Therapeutic agents – drugs(over doses, unusual adverse effects, frequent administrations of therapeutic doses & drug interactions)
 - Industrial chemicals - cause environmental pollution & direct hazard.
 - Household chemicals - cleaning agents, cosmetics & personal products & berries.
 - Environmental contaminants - industrial processes, pesticides &smokes from factories & vehicles.
 - Natural toxicants - toxin from plants and animals for both defense & offensive purposes. Via containing in food, by accidental ingestions of poisonous plants or animals & by stinging & biting.
 - Food additives – have usually low biological activity & added to alter the flavor or color, prevent spoilage or in some other way change the nature of the food stuff.
 - Traditional medicines (Botanicals) – in their natural & unprocessed form, they were used as Medicine. Can be adulterated, misbranded or contaminated. Furthermore, the doses for active botanical substances may be higher.

- Drugs of abuse - Excessive or improper use of drugs or other substances for non-medical purposes (altering consciousness, for body building). High potential of dependence & tolerance (e.g. alcohol, nicotine...)

- **Routes and risks, management of heavy metals poisoning**

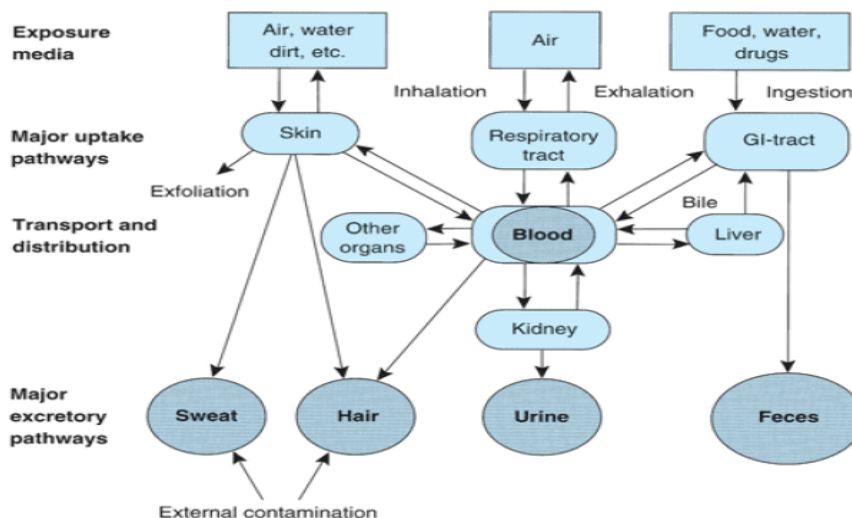
- Although most metals occur in nature in rocks, ores, soil, water, and air, levels are usually low and widely dispersed.
- Known essential metals include: cobalt, copper, iron, magnesium, manganese, zinc, selenium and molybdenum. They are toxic when excess.
- Major toxic metals include - arsenic, cadmium, chromium, mercury, lead, iron.
- For a metal to exert its toxicity, it must cross the membrane and enter the cell.
- If the metal is in a lipid-soluble form such as methyl mercury, it readily penetrates the membrane.
- When bound to proteins such as cadmium - metallothionein, the metal is taken into the cell by endocytosis.
- Other metals (e.g., lead) may be absorbed by passive diffusion.
- The toxic effects of metals usually involve interaction between the free metal and the cellular target. These targets tend to be specific biochemical processes and/or cellular and subcellular membranes.
- Reacts in the body with ligands containing oxygen (-OH, - COO⁻, - OPO₃H⁻, >CO), sulfur (- SH, - S-S-) and nitrogen (-NH₂, >NH) by forming stable co-ordination complex.

Pathophysiological actions of non-essential elements

- Substituting essential elements in enzymes, neurotransmission, structural components
- Reacting covalently or non covalently with enzymes, membrane, DNA
- Stimulate production of reactive oxygen species

Route of exposure

- Inhalation , ingestion and through skin



Risk factors for poisoning

- ✓ Working in agriculture
- ✓ Pharmaceutical manufacturing and industrial work
- Biomarkers are used to measure exposure, toxicity, and susceptibility to toxic metals. Biomarkers of exposure, also called biological monitors, such as metal concentrations in blood, and urine.
- A critical determinant of retention of a metal is its biological half-life, varies according to the metal as well as the organ or tissue; in the case of many metals, more than one half-life is needed to fully describe the retention time.
- Blood, urine, and hair are the most accessible tissues in which to measure an exposure or dose; they are sometimes referred to as *indicator tissues*.
- Blood and urine concentrations usually reflect recent exposure and correlate best with acute effects. An exception is urinary cadmium, which may reflect renal damage related to an accumulation of cadmium in the kidney.
- Partitioning of metal between cells and plasma and between filterable and non-filterable components of plasma should provide more precise information regarding the presence of biologically active forms of a particular metal.
- Speciation of toxic metals in urine may also provide diagnostic insights. For example, cadmium metallothionein in plasma or urine may be of greater toxicologic significance than total cadmium.
- Hair might be useful in assessing variations in exposure to metals over the long term. Hair levels of mercury have been found to be a reliable measure of exposure to alkyl or methyl mercury. For most metals, however, hair is not a reliable tissue b/c of external contamination.

Factors Influencing Toxicity of Metals

- Interactions with essential metals (Pb and Ca^{2+} and Fe; and Cd and Fe) - if metabolism is similar
- Formation of metal-protein complexes - for detoxification, but no enzymatic activity
- Age and stage of development - children and elders
- Lifestyle factors - smoking and alcohol
- Chemical form or speciation
- Immune status of host - metals that provoke immune reactions include mercury, gold, platinum, beryllium, chromium, and nickel.

Specific metal binding proteins

- Metallothioneins - for essential metal homeostasis and protection against metal toxicity through thiol binding (Cd, Cu, Hg, Ag, and Zn. In)
 - ✓ Important in regulation of Cu and particularly Zn metabolism
 - ✓ Their induction is highly stimulated by Zn; Cd is strong inducer
- Transferrin – bind ferric iron in plasma. Transport of iron across cell membranes occurs by receptor-mediated endocytosis of ferric transferrin. Also transports Al^{+3} and Mn^{+2} .

- Ferritin - storage protein for iron in the reticulo-endothelial cells of liver, spleen, and bone. It plays a major role in hepatic turnover of iron. Serve as a general metal detoxicant, since it binds a variety of toxic metals including Cd, Zn, Be and Al.
- Ceruloplasmin - copper-containing glycoprotein oxidase in plasma that converts ferrous to ferric iron. This protein also stimulates iron uptake by a transferrin in dependent mechanism

Membrane carrier proteins

- Some metals transported in the free ionic forms, as through calcium channels, many metals are transported as complexes with endogenous ligands on transport systems intended for the ligand itself.
 - Phosphate and sulfate transporters can carry metal oxyanions across the plasma membrane. Vanadate and arsenate are structurally similar to phosphate and can compete with phosphate for transport as well as intracellular binding sites; chromate, molybdate, and selenate are structurally similar to sulfate and are carried on the sulfate transporter.
 - Amino acid and peptide transporters and organic solute carriers also accept metals as complexes with endogenous molecules such as amino acids, peptides, and bicarbonate (copper, zinc, lead, methyl mercury).

Arsenic

- The levels in air and water are low, except in some areas and the major source of human exposure is food. Arsenic compounds are used as pesticides, glassware, paints, and pigments. Only a small amount crosses the blood – brain barrier. Arsenic may also replace phosphorus in bone tissue and be stored for years.
- Arsenic compounds can be inorganic or organic (w/c is methylated) but generally occur in three forms:
 1. **Pentavalent, As +5** - organic or arsenate compounds (e.g., alkyl arsenates);
 - uncoupling mitochondrial oxidative phosphorylation (Competition with phosphate during oxidative phosphorylation & Inhibition of energy-linked reduction of NAD)
 2. **Trivalent, As +3** - inorganic or arsenate compounds (e.g., sodium arsenate, arsenic trioxide); lipid soluble; react with – SH group(pyruvate dehydrogenase system is sensitive and inhibits succinic dehydrogenase activity)
 3. **Arsine gas, AsH₃** - a colorless gas formed by the action of acids on arsenic.
 - most toxic form
 - used in the semiconductor industry
 - potent hemolytic effect associated with alteration of ion flux across the erythrocyte membrane, disrupting Na⁺/K⁺ pump.
- Toxicity - Organic arsenicals (dimethyl As < methyl As) < As⁵⁺ < As³⁺ < arsine (AsH₃)
The toxicity of a given arsenical is related to the rate of its clearance from the body and therefore to its degree of accumulation in tissues.
- **Toxic effects** - GI toxicity, hematotoxicity, hepatotoxicity, neurotoxicity, nephrotoxicity, cardiotoxicity + proven to be carcinogen (particularly skin and lung but other organs involved).
- **Acute** - severe GI symptoms occur: vomiting, profuse diarrhea (rice – water stools), severe abdominal pain; vasodilatation, myocardial depression, cerebral edema, and

distal peripheral neuropathy may also follow; later jaundice and renal failure; death usually results from circulatory failure.

- **Chronic** - nonspecific symptoms such as diarrhea, abdominal pain, hyperpigmentation, & hyperkeratosis. Symmetrical sensory neuropathy often follows. Late changes include gangrene of the extremities (Blackfoot disease), anemia, and cancer.
- Concentrate in keratin (hair & nail aka “mee’s lines”), garlic odor on breath due to arsine gas.
- **Carcinogenicity** - skin cancer (BCC & SCC), lung carcinoma, bladder cancer.
- **Management** –
 - ✓ Supportive/ symptomatic therapy – treat hypotension, fluid loss
 - ✓ Remove sources
 - ✓ GI decontamination: Ipecac, lavage
 - ✓ Chelation therapy (Dimercaprol, Penicillamine, Succimer)

Cadmium

- Used as a pigment in paints and plastics, in electroplating, and in making alloys and alkali storage batteries (e.g., nickel - cadmium batteries). Grains, cereal products, and leafy vegetables usually constitute the main source of cadmium in food.
- Cadmium is present in the circulatory system-before arrival in liver it's bound to blood cells and albumin) while bound to the metal - binding protein, MT in liver, then either stored in liver or CdMT is reabsorbed by the proximal tubule cells and accumulates within the lysosomes. Subsequent degradation of the CdMT complex releases Cd +2 , which inhibits lysosomal function, resulting in cell injury.
- **Toxicity**- Slowly excreted from the body, with a half - life of about 30 years.
- **Acute** - from local irritation i.e. ingestion- nausea, vomiting, and abdominal pain.
 - Inhalation- pulmonary edema and chemical pneumonitis.
- **Chronic** - COPD, chronic renal tubular disease, effects on CVS and skeletal muscle.
- **Carcinogenicity** - lung cancer & prostate cancer
- **Management** - no chelation therapy but experimentally, DMSA and CaEDTA best reduce acute mortality in combination with glutathione.

Chromium

- Used in making stainless steel, various alloys, and pigments. The levels of this metal are generally very low in air, water, and food, and the major source of human exposure is occupational.
- Chromium occurs in a number of oxidation states from Cr +2 to Cr +6 , but the important ones are:
 - Trivalent(Cr +3) - in nature and
 - Hexavalent (Cr +6) - in industries, not water soluble & more readily absorbed across cell membranes
- **Toxicity** - the hexavalent form is reduced to the trivalent form, which can complex with intracellular macromolecules, resulting in toxicity.
- **Acute** - tubular and glomerular damage.

- **Chronic** - ulceration and perforation of the nasal septum, ulceration of skin, asthma.
- **Carcinogenicity** - cancer of the respiratory tract b/c of reduction to Cr^{+3} and generation of reactive intermediates, leading to bronchogenic carcinoma.

Mercury

- Mercury exists in the environment in three main chemical forms:
 - 1) elemental (Hg^0) - 80% absorption if inhaled but less GIT absorption, cross BBB
 - exposure is occupational
 - neurotoxicity and nephrotoxicity
 - 2) inorganic salts-mercurous (Hg^{+}) - less toxic and less absorption
 - mercuric (Hg^{2+}) - more toxic
 - do not readily pass the BBB or the placenta
 - nephrotoxicity
 - 3) organic-methylmercury and dimethyl mercury
 - Cross the BBB or the placenta – more neurotoxic & teratogenic, hepatotoxicity.
- Organic mercury (methyl mercury ~90%) $\gg \text{Hg}^{2+} > \text{Hg}^{+} > \text{Hg}^0$ GIT absorption
- After absorption, all(Hg^0 , Hg^{+} , methyl mercury) $\rightarrow \text{Hg}^{2+}$
- Mercurials have high affinity for thiols; also combines with phosphoryl, carboxyl, amide, and amine groups
- **Management**
 - Elemental Mercury Vapor: termination of exposure, chelation therapy
 - Acute ingestion of inorganic salt: GI decontamination, electrolyte and fluid Replacement to treat shock
- **Chelation therapy:** important for acute and chronic poisoning (less effective for alkyl mercury)
 - : Dimercaprol (BAL), penicillamine, N-acetylcysteine

Lead

- Exposure may be through air, water, or food sources. Used in batteries, metal alloys, glass, plastics and ceramics. It exists in two forms :
 - Organic-tetraethyl lead (TEL)
 - respiratory, GIT, skin
 - Inorganic-oxides and salts
 - GIT (more absorption for children), inhalation, less absorption via skin
 - crosses the placenta, and in children pass BBB.
- Lead is transported across the GI mucosa using same transporter mechanism to that of calcium – competition.
- Low dietary calcium, iron deficiency, and empty stomach – increase lead absorption
- Initially, lead is distributed in the blood, liver, and kidney; after prolonged exposure, as much as 95% of the body burden of lead is found in bone tissue
- **Toxicity:**
- **Acute**-sweet metallic taste, salivation, intestinal colic
- **Chronic**-
 - ✓Hematologic: basophilic stippling, hypochromic microcytic anemia

- Inhibition of enzymes of heme synthesis (ALDA-result in urine aminolevulinic acid & heme synthase).
- Shorten RBCs life span through inhibition of sodium- and potassium-dependent ATPases (fragile cell membrane).
- ✓ Neurologic (lead encephalopathy): ataxia, nausea, vomiting, restlessness, irritability, learning disability and regression, convulsions, coma
- Interfere neurotransmitter function, disrupt Ca^{+2} metabolism and damage membrane
- ✓ GI (lead colic): anorexia, constipation, metallic taste, colicky abdominal pain
- ✓ Neuromuscular (lead palsy): wrist-drop, foot – drop, fatigue, muscle weakness
- ✓ Renal: Fanconi-like syndrome (reversible), chronic nephritis (irreversible)
- Others: GIT effects, reproductive toxicity (sterility and neonatal deaths), CVS (hypertension).
- **Management**
 - Prevention of further exposure
 - Seizures with diazepam or phenytoin
 - Fluid and electrolyte maintenance
 - Cerebral edema with mannitol and dexamethasone
- **Chelation therapy:** is indicated in symptomatic patients or in patients with a blood lead concentration in excess of 50 to 60 $\mu\text{g/dl}$
- Chelators in use are : CaNa_2EDTA and dimercaprol (for lead encephalopathy), D-penicillamine, and succimer

Iron

- Regulation involves intake, stores, and loss.
- Absorption occurs in two steps:
 - (1) Absorption of ferrous ions from the intestinal lumen into the mucosal cells and
 - (2) from the mucosal cell to plasma, transfer to storage sites.
- Transported by plasma protein transferrin & stored in hemoglobin, myoglobin, enzymes, ferritin and hemosiderin.
- Excreted, and some contained in shed intestinal cells, bile, urine, sweat, nails, and hair.
- **Toxicity-**
 - Acute mucosal cell damage & capillary endothelial cell damage in liver
 - Fe accumulation in Kupffer cells toxic to hepatocyte mitochondria ---
- **Iron deficiency**
- Acute exposures- in children:
 - Early: vomiting, shock and metabolic acidosis, liver damage, coagulation defects.
 - Late: renal failure and hepatic cirrhosis. The mechanism of the toxicity
- Chronic iron overload due to idiopathic hemochromatosis (abnormal absorption), excess dietary iron or frequent blood transfusions (transfusional siderosis).
- **Management**
 - Symptomatic treatment
 - ✓ For CV – collapse and dehydration --- IV fluid
 - ✓ Metabolic acidosis--- Sodium bicarbonate
 - ✓ Convulsions --- barbiturates
 - Prevent continued absorption: vomiting (ipecac) or gastric lavage complexation (desferrioxamine, phosphates, bicarbonate)
 - Chelation therapy with desferrioxamine (IV, IM)

- **The common pesticides poisoning and management**

How are you exposed to pesticide?

- Inadvertent ingestion- solution for home use
- Dermal and respiratory exposure

1. Organophosphate (OP)

- similar mechanism as nerve gas(sarin, tabun, soman)
- **Mechanism of toxicity**
- Irreversible inhibitor of acetylcholinestrerase via phosphorylation this can increase the concentration acetylcholine
- Acetylcholinestrerase undergo aging, no spontaneous regeneration of the enzyme, these is toxic manifestation of the OP.
- **Sign and symptom of OP poisoning**
- The sign and symptoms are due to excess acetylcholine
 - Salivation
 - Lacrimation
 - Miosis
 - Hypertension
 - Bronchoconstriction
- Excess depolarization at nicotinic neuromuscular synapse leads to muscle weakness and paralysis.
- CNS cholinergic stimulation suppress the medullary center for respiration and cardiovascular system, this is the major cause of death.
- **Management**
 - Antidote
- atropine which is muscarinic blocker
- Pralidoxime which break the covalent bond between OP and acetylcholinestrerase-remember pralidoxime do not work if aging occur

2. Carbamates

- Mechanism of toxicity=> Reversible inhibitor of acetylcholinestrerase via carbamoylation
- Carbamates are not lipid soluble- so do not cross BBB.
- Sign and symptom of toxicity=> the same as OP poisoning but much less severe and has no CNS effect.
- Management:
 - Atropine as an antidote - remember pralidoxime is not useful here since there is no covalent bond.

3. ORGANOCHLORINE (OC)

- E.g. Lindane, DDT...
- Are lipid soluble compounds; enter BBB and also pass through skin.

- Mechanism of toxicity:
 - Interfere with the neurotransmitter GABA
 - Inhibit GABAergic neuron.
- **Sign and symptom of toxicity**
- Most of the toxic effects of OCs are due to CNS stimulation since GABAergic neurons are inhibited
- Apprehension, irritability, confusion, sensory disturbances, dizziness, tremors, and seizures.
- myocardial arrhythmias, respiratory failure
- **CLINICAL MANAGEMENT OF ACUTE POISONING:**
- symptomatic and supportive management of the condition
- use lidocaine for cardiac arrhythmia and benzodiazepine for seizure

● **Risk factors for house hold products poisoning**

- The ingestion of household products by a child is the most common pediatric medical emergency.
- Most house hold product have relatively low toxicity
- The top household products ingested are
 - cleaning agents,
 - cosmetics
 - personal products & berries.
 - Bleach is the most common source of poisoning
- Sign and symptom of toxicity:
 - burning to the throat, mouth & esophagus.
 - The tissue damage results in edema in the pharynx & larynx.

● **AIR POLLUTANTS**

- The major air pollutants are, CO (which accounts for about 50% of the total amount of air pollutants), sulfur oxides (18%), hydrocarbons (12%), particulate matter (e.g., smoke particles, 10%), and nitrogen oxides (6%)
- Air pollution appears to be a contributing factor in bronchitis, obstructive pulmonary disease, and lung cancer

Carbon monoxide

- Odorless, colorless gas that competes avidly with oxygen for hemoglobin. The affinity of CO for hemoglobin is more than 200-fold greater than that of oxygen. It causes tissue hypoxia. Headache occurs first, followed by confusion, decreased visual acuity, tachycardia, syncope, coma, seizures, and death.
- Treatment is removal of the source of CO and 100% oxygen. Hyperbaric oxygen accelerates the clearance of carbon monoxide.

Sulfur dioxide

- Colorless, irritating gas formed from the combustion of fossil fuels. Conjunctival and bronchial irritation (especially in individuals with asthma) is the primary signs of exposure. Forming sulfurous acid when in contact with mucous membrane is the mechanism. Removal from exposure to SO₂ and relief of irritation and inflammation constitutes the major treatment.

Nitrogen oxides

- Nitrogen dioxide (NO₂), a brownish irritant gas, is the principal member of this group. It is formed in fires and in silage on farms. NO₂ causes deep lung irritation and pulmonary edema. Irritation of the eyes, nose, and throat is common. No specific treatment is available. Measures to reduce inflammation and pulmonary edema are important.

Ozone

- Is a bluish irritant gas produced in air and water purification devices and in electrical fields. Exposure to 0.01–0.1 ppm may cause irritation and dryness of the mucous membranes. Pulmonary function may be impaired at higher concentrations. Chronic exposure leads to bronchitis, bronchiolitis, pulmonary fibrosis, and emphysema. No specific treatment is available. Measures that reduce inflammation and pulmonary edema are emphasized.

● MANAGEMENT OF ACID AND ALKALI POISONING

- Acids (hydrogen ions) desiccate tissues by drawing water out of cells to form hydronium ions (H₃O⁺), which results in coagulative necrosis and eschar formation. Depending upon the severity of the exposure, eschars can prevent the deeper penetration of acids, causing superficial damage to the tissues. Severe acidic burns can penetrate the self-limited tissue barriers and cause more extensive damage. Alkalis cause saponification of membranes, which allows for rapid and extensive burns through tissues.
- Clinical presentation of an acid ingestion-
 - 1. Upper respiratory – stridor, drooling, respiratory distress
 - 2. GI – oropharyngeal pain and burns, odynophagia, nausea, vomiting, abdominal/chest/epigastric pain
 - 3. Severe injury can manifest with hypotension, metabolic acidosis, DIC, mediastinitis
- Acid ingestion management-
 - 1. Supportive care – give water or milk to dilute, especially with powder or granular preparations. It is important to make sure the patient can swallow without difficulty.
 - 2. Decontamination with an NG tube and gentle suction is controversial.
 - 3. Endoscopic evaluation as needed
 - 4. Emergent laparotomy for suspected perforation

- 5. AVOID cathartics, neutralization, induced-emesis, activated charcoal, steroids
- 6. Hydrofluoric acid burns are managed differently-HF is a weak acid that easily crosses membranes and causes deeper tissue damage. Fluoride ions chelate calcium and magnesium and form deposits. Dysrhythmias may develop secondary to hypocalcemia or hypomagnesaemia. Treatment for superficial injuries involves topical calcium gluconate gel. Deeper wounds with systemic symptoms may require intra-arterial calcium gluconate with additional IV calcium and magnesium replacement
- 7. Ocular acid injuries should be managed - Copious irrigation with normal saline is the mainstay of therapy. Several liters of irrigation are usually needed to raise the pH back to 7.40. Fluorescein exam should be done to confirm that a normal pH is achieved. An ophthalmology consult is warranted for corneal involvement.
- Alkali poisonings can be managed by the above 5 basic mechanisms in case of ocular injuries due to extensive penetration, alkali burns may require extensive irrigation.

● COMMON FOOD CONTAMINANTS AND MANAGEMENT

- Bacteria producing performed toxins include:- Staphylococcus aureus, Clostridium perfringens, Bacillus cereus, Clostridium botulinum
 - Performed toxins are difficult to eradicate because they are heat resistant(not easily deactivated by cooking/boiling)
 - IP is short and symptoms appear within 6-12 hours after consumption
- Bacteria producing toxin in the gut include:- Bacillus cereus, Campylobacter jejuni, Enterotoxigenic and enterohemorrhagic Escherichia coli, Vibrio parahemolyticus, Clostridium perfringens
- Finding abundant WBC's on a stool smear may differentiate a bacterial etiology from other causes of food poisoning.
- **AFLATOXIN**- mycotoxin from Aspergillus flavus. Occurs in susceptible crops as mixtures of aflatoxins B1, B2, G1, and G2, with only aflatoxins B1 and G1 demonstrating carcinogenicity.
- For more information see microbiology part.

● PREGNANT DRUG CLASSIFICATION AND REPRODUCTIVE TOXICANTS

- To prevent adverse events to fetus USFDA(1979) classify drugs under five categories

PREGNANCY CATEGORY A

- Controlled studies show no risk to the fetus
- Adequate well controlled studies in pregnant women have demonstrated no risk to the fetus

PREGNANCY CATEGORY B

- There is no evidence of risk in humans

- Animal studies show no risk ,but human findings do not
- E.g.: -methyldopa

PREGNANCY CATEGORY C

- Risk cannot be ruled out.
- Human studies are lacking, and animal studies are either positive for fetal risk or lacking
- The drug may be used during pregnancy if the potential benefits outweigh its possible risk
- E.g.: -furosemide

PREGNANCY CATEGORY D

- There is positive evidence of risk to the human fetus
- Investigational or post marketing data show risk to the fetus
- However, potential benefits may outweigh the risk to the fetus. If needed in a life threatening situation or a serious disease, the drug may be acceptable if safer drugs cannot be used or are ineffective
- E.g.: -phenytoin

PREGNANCY CATEGORY X

- Use of the drug is contraindicated in pregnancy
- Studies in animals or humans or investigational or post marketing reports have shown fetal risk that clearly outweighs any possible benefit to the patient.
- E.g.: -warfarin, retinoids (avoid pregnancy for 2 years after use)

(Microbiology Part)

● UTI and lab diagnosis

- The UUT includes the ureters and the kidney while the LUT includes urethra and urinary bladder.
- UTI is very common and mostly occur as a result of urethral infection by organisms in the colon.
- It is common in sexually active women. Its rate increases in post menopausal women(bc they experience bladder and uterine prolapse plus the decrease in estrogen will result in change in normal flora i.e. loss of lacto bacilli .the rate is high in older men with prostatic hyperplasia(blockage of urine will lead to storage of urine which leads to infection)
- UTI-is defined as a significant bacteruria and WBC with other related problems like dysuria, increased urinary frequency.....

Common terms

- ☐ Bacteriuria: the presence of bacteria in the urine

- ❑ Asymptomatic bacteriuria (ASB): presence of bacteria in the urine in the absence of symptoms/signs of infection.
- ❑ Pyuria: the occurrence of WBCs in the urine
- ❑ Recurrent: >3 symptomatic UTIs within 12 months following clinical therapy.
- ❑ Re-infection: recurrent UTI caused by a different pathogen at any time.
- ❑ Relapse: recurrent UTI caused by same species causing original UTI within two weeks after therapy.
- ❑ Uncomplicated UTI – Infection in a structurally and neurologically normal urinary tract.
 - Simple cystitis of short (1-5 day) duration
- ❑ Complicated UTI – Infection in a urinary tract with functional or structural abnormalities (example: - indwelling catheters and renal calculi).

E.g. Cystitis of long duration or hemorrhagic cystitis

Etiology

- Most common etiologic agents are aerobic members of fecal matter.
- From these G-ve rods take the lion's share.
- And from these G-ve rods E.coli is the most common one.

1) E.coli

- The most common cause.(>80% of acute and 40-50% of chronic infections)
- It breaks down lactose rapidly
- Has complex Ags based on O, K, and H antigens
- It is a normal flora in the GIT
- Causes different infections including diarrhea, meningitis, UTI....when it acquires different virulence factors.
- mostly by UPEC(uropathogenic E.coli) which can attach specifically to receptors of the renal pelvis mucosa with pyelonephritis-associated pilli (PAP, P fimbriae)(ty 1 pilli) or non-fimbrial adhesins (NFA)(produces hemolysin (HlyA)

Diagnosis: gram stain: -show pink (red) rod shaped bacteria

Culture: the preferred one, culture on endo agar.

Show pink color on MacConkey agar since they ferment lactose rapidly.

2) Genus klebsiella

- Non motile, capsulated, lactose fermenting
- K. pneumonia - (most common type), commensal in the intestine
- common nosocomial pathogen

Diagnosis: gram stain: - red rods

Culture: on MacConkey agar-large, mucoid pink colonies.

Serology: capsular polysaccharide

- Note that non motility and presence of capsular polysaccharide is the main distinguishing factor of klebsiella

3) Genus Proteus

- Very known for its motility
- Non encapsulated, nosocomial pathogens
- Have urease which splits urea to ammonia and CO₂
- Since NH₃ increases urine PH, they are known to form calculi (stone)
- P. mirabilis and P. vulgaris are the medically important species
- Associated with long term catheterization (CAUTI) and UT abnormalities.
- Diagnosis: gram stain: rods

Culture: swarming growth on blood agar

- Cross react with Weil-felix test (remember the 3 strains OX-19, OX-2, and OX-K are used to determine the presence of rickettsia because they have cross reacting Ags with rickettsia)

4) Genus *Pseudomonas*

- Motile, n.lactose fermenter
 - *P. aeruginosa* is the medically important species
 - Causes UTI only in people with weakened immune system. (Eg: neutropenic patients)
 - nosocomial pathogen
 - Produce two pigments, i.e., both yellow-green fluorescein and blue-green pyocyanin
 - usually enters body through injuries
 - resistant to many antibiotics (due to the architecture of its membrane)
 - Exotoxin, exoenzyme and cytotoxin are its virulence factors.
 - Diagnosis: gram stain: red rods
- Culture: obligate aerobe

Gram positive

1) *Staphylococcus saprophyticus*

- Catalase +ve and coagulase -ve
- Normal flora of the female genital tract and perineum
- Sexual intercourse increases the risk since the bacteria are displaced to the urethra.
- Most cases occur within 24 hours of sex, earning this infection the nickname "honeymoon cysti"
- Symptoms of cysti include burning sensation, dripping effect, razor pain during intercourse....
- Diagnosis: novobiocin to distinguish it from *S.epidermidis*.

Pathogenesis of UTI

- 4 routes of spread:-
 - A) Hematogenous route
 - B) Ascending route
 - C) Lymphatogenous route
 - D) Direct extension from other organs
- A and B are the most common routes
 - A) Hematogenous**
 - Less common compared to ascending route.
 - In people with endocarditis or bacteraemia caused by the gram positive *S.aureus*.
 - The kidney is the most common site.
 - It's rare in gram -ves.
 - B) Ascending**
 - The most common cause.

- Ascension of organisms up the urethra to the bladder
- Whether infection develops depends upon the:
 - particular organism
 - size of the inoculum
 - adequacy of host defenses.

Host factors

- shorter urethra and proximity to anus
- urethral massage that accompanies sexual intercourse
- diaphragm use - manipulation to place it may facilitate colonization
- Spermicidal use raises PH, killing lactobacilli and facilitating.
- women with P1 blood group have epithelial cell receptors that mediate attachment of bacteria
- Hygiene, vesticulourethral reflex....
- All these host factors are the reason why women are more infected than men.

Bacterial factors

- Presence of virulence factors
- Enhanced adherence to uroepithelium..... (e.g. E.coli has p and type 1 fimbriae; although type- 1 is blocked by mannose the p fimbriae is resistant)
- Due to this, it uses the p fimb rather than the ty 1 after reaching to the kidney to facilitate attachment to the parenchyma.
- In general:
 - adhesions
 - Toxins
 - Invasiveness are the main bacterial factors.

Specific UTI infections

1) Urethritis

- infection of urethra
- mostly by ST(sexually transmitted) organisms
- Characteristics: dysuria and urethral discharge in men.

2) Cystitis

- infection of urinary bladder
- Usually after sexual intercourse in women (honey moon cystitis) and if it's recurrent due to chronic bacterial prostatitis in men.

- Characteristics: urgency and frequency of urination, cloudy (turbid) urine + symptoms of urethritis.

3) Acute urethral syndrome

- occurs in women
- also called dysuria-pyuria syndrome (pain and pus in the urine)
- Resembles cystitis but shows lower colony counts than cystitis.

4) Asymptomatic bacteriuria

UTI-symptoms

- + or – pyuria

5) Acute pyelonephritis

- Infection of kidney parenchyma
- Causes about 20% of community acquired bacteremia.
- Characteristics: symptoms of urethritis and cystitis+fever, chills, nausea, costovertebral angle tenderness.
- But in children less than 2 yrs of age fever, emesis, and failure to gain weight are the usual symptoms.

Fungal UTI

- Candida species is the major one. Characterized by candiduria.
- Mainly seen in diabetic patients and immune compromised people.
- If candiduria persists after 48-72 hrs after catheter removal it's also a true candidiasis.
- Can be in: lower UT - in patients with urinary catheters
Kidney - usually hematogenously from GIT and sometimes by ascending infection.
- Other invasive fungi like Aspergillus sp, Mucoraceae sp, Histoplasma can also cause UTI.

General diagnosis of UTI

- Microscopic examination of urine should be the first step.
- result proven only by culture
- >10⁵ bacteria/ml- means there is infection
- more than 10 leukocytes/ μ l- means there is infection
- Hematuria and proteinuria - indicate that uncomplicated cystitis has turned in to a complicated one.

GOOD LUCK!!!

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